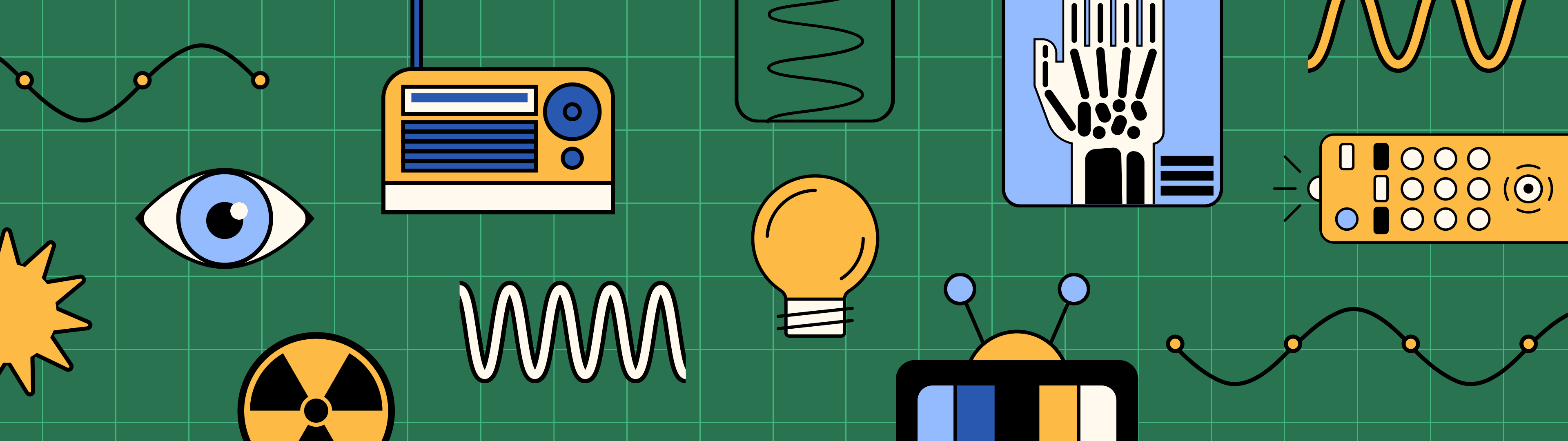
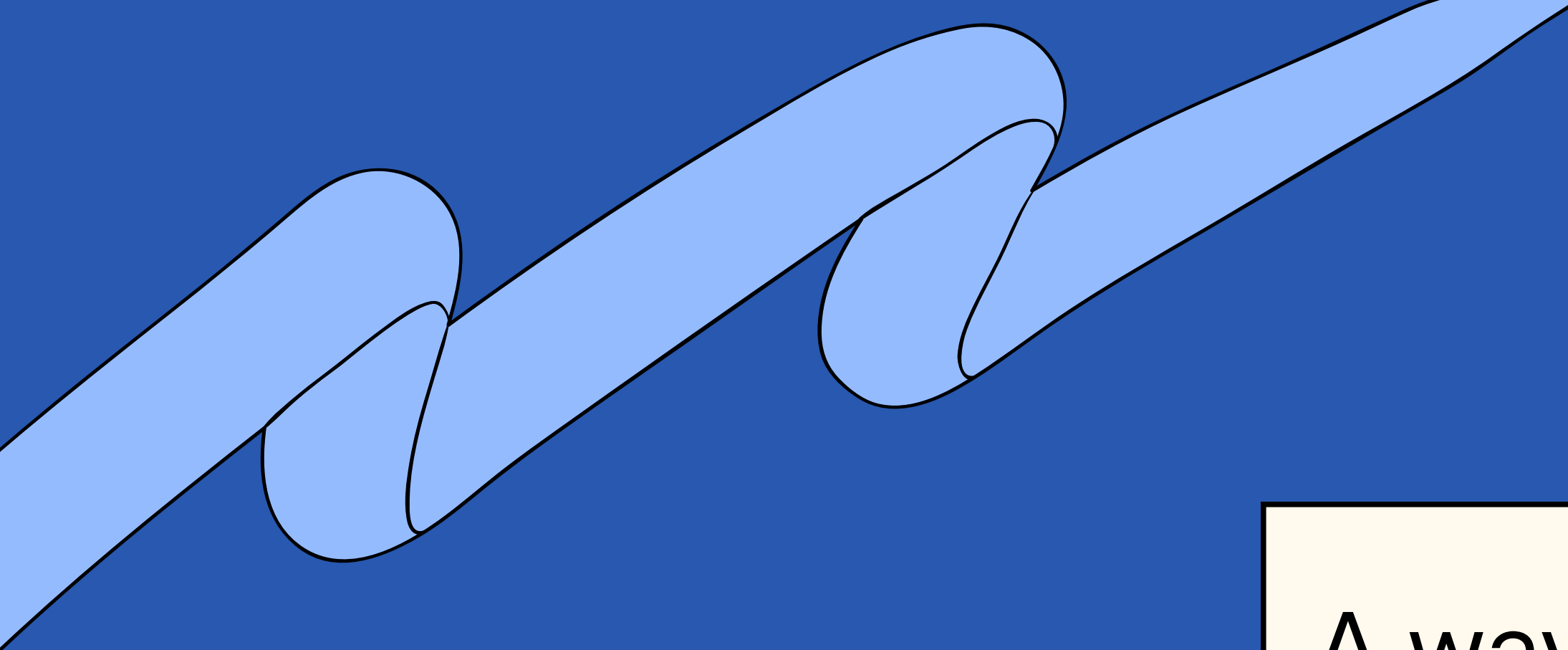




Waves

& Wave Properties

Exploring the types of waves and their properties



Waves

A wave is a disturbance that transfers energy through a medium or space.

Waves do not transfer matter — only energy.

Types of Waves

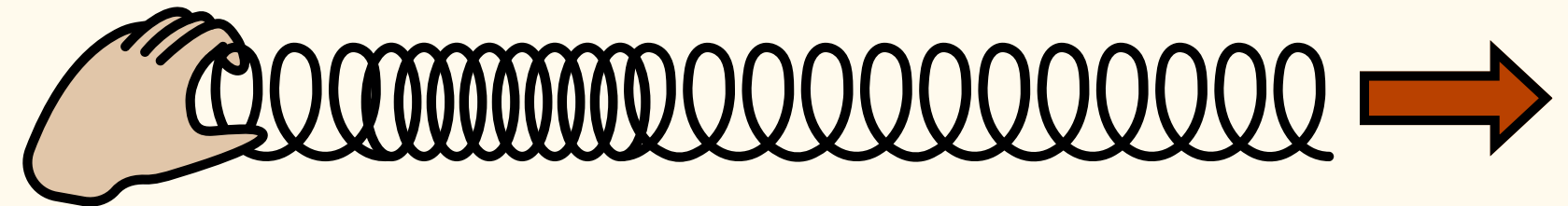
**according to
Direction of
Vibration**

Longitudinal Waves

direction of vibration

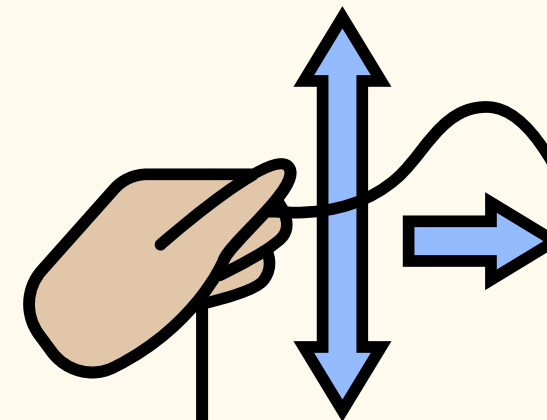


direction of wave
propagation

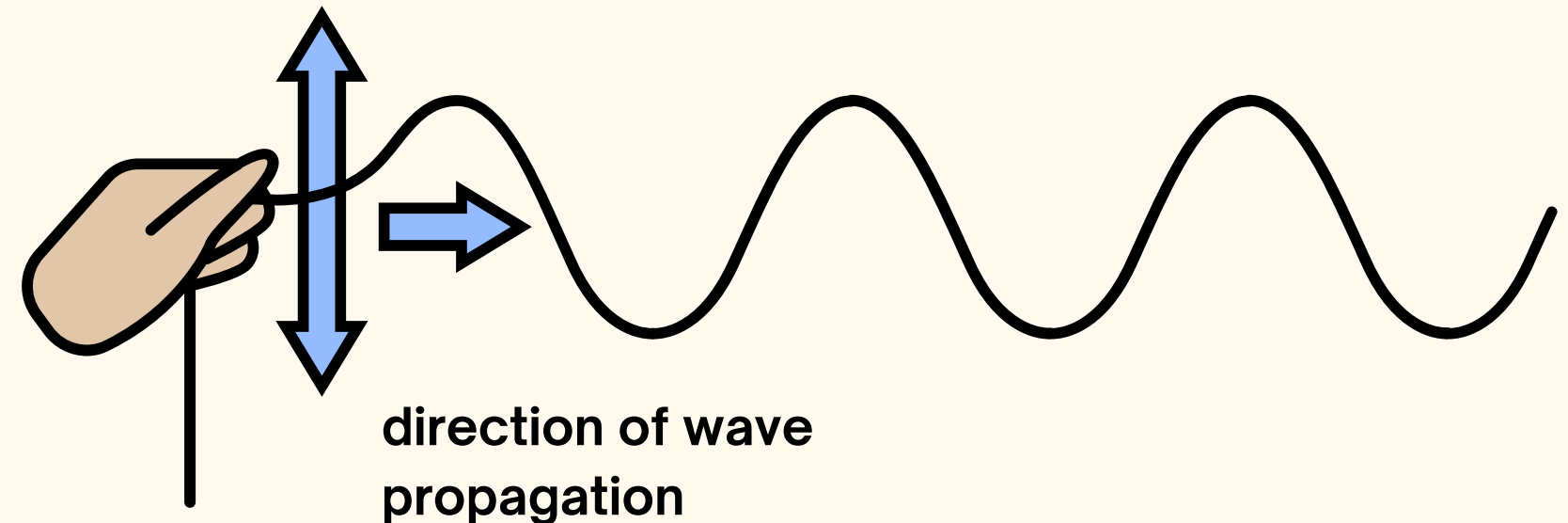


Transverse Waves

direction of vibration



direction of wave
propagation



Longitudinal Waves

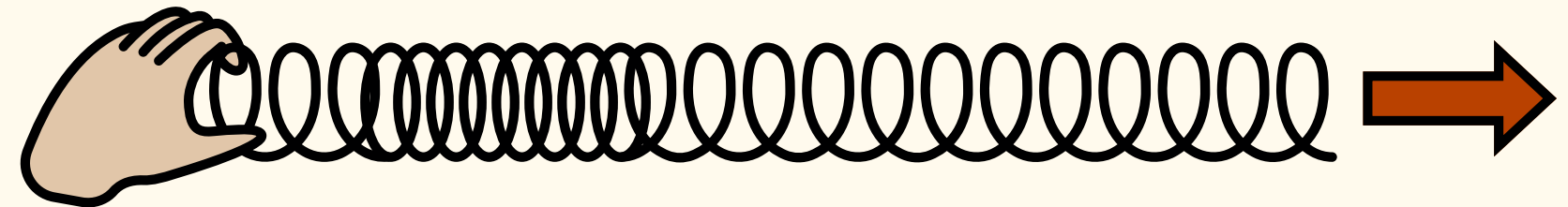
Longitudinal Waves

- Vibrations occur **parallel** to the direction the wave travels
- Have regions of **compression** and **rarefaction**
- Sound is an example of a longitudinal wave

direction of vibration



direction of wave propagation



compression

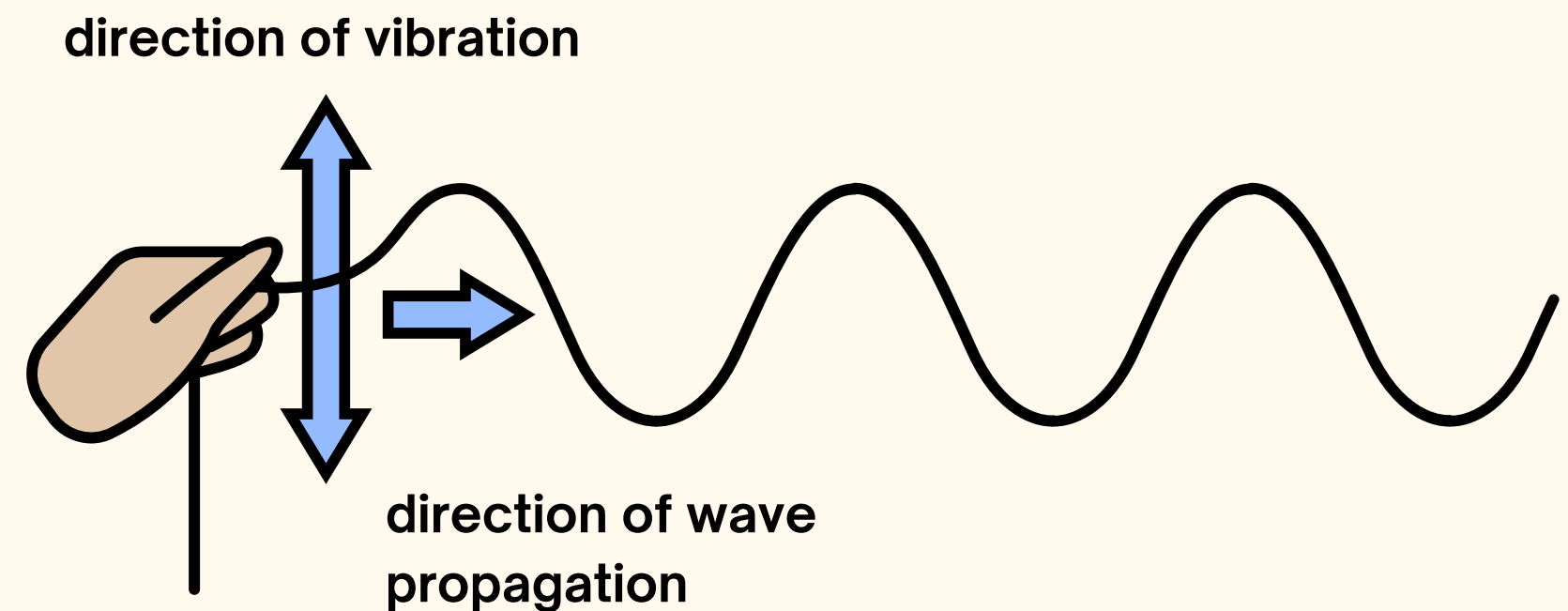
rarefaction

PUSHING A SLINKY BACKWARDS AND FORWARDS
PRODUCES A LONGITUDINAL WAVE.

Transverse Waves

Transverse Waves

- Vibrations occur perpendicular to the direction the wave travels
- Have **crests** and **troughs**
- Light is an example of a transverse wave



SHAKING A ROPE UP AND DOWN PRODUCES A TRANSVERSE WAVE.

Parts of a Transverse Wave

Crest

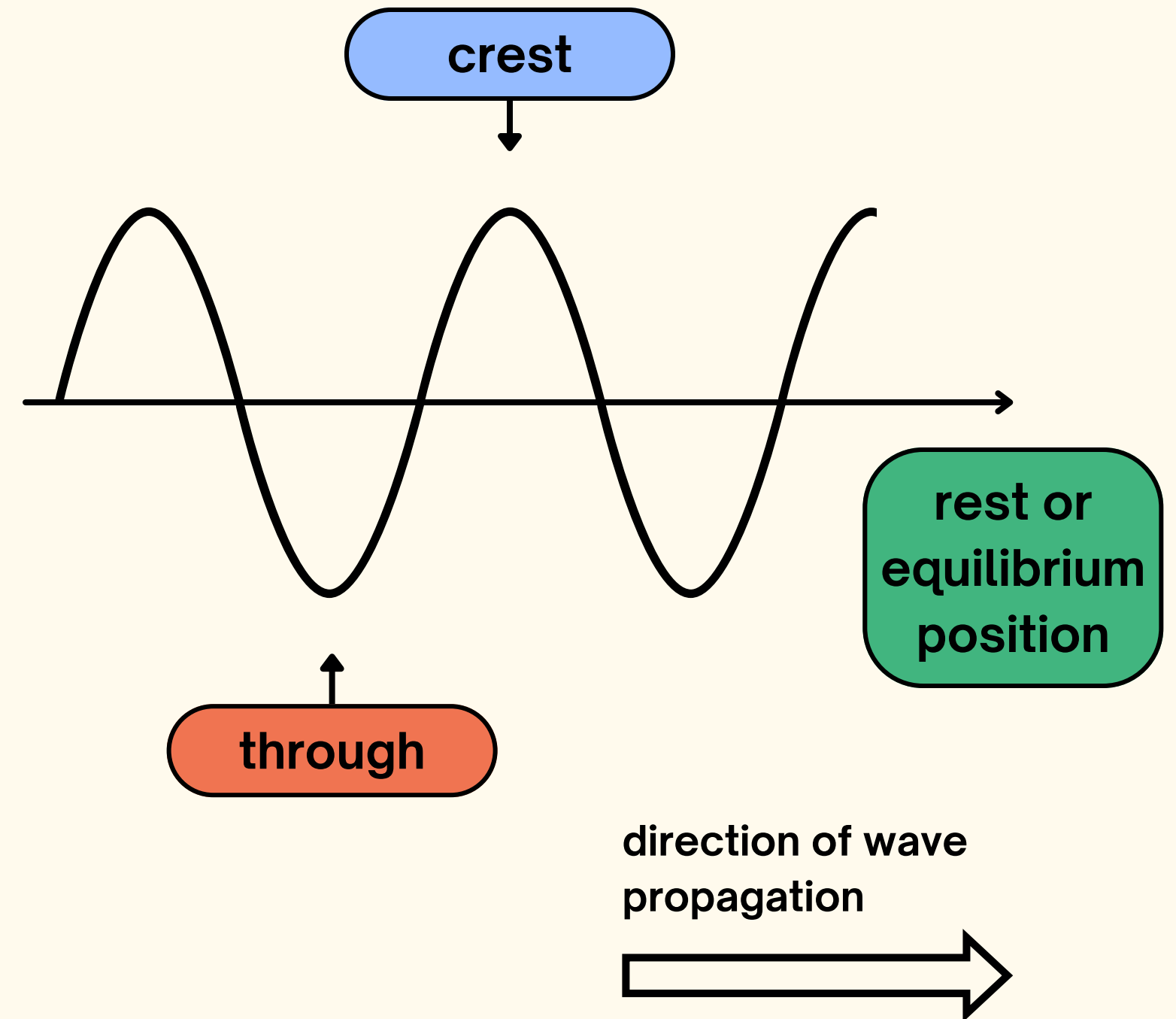
Otherwise known as the peak, the highest point of a wave

Rest or Equilibrium Position

The undisturbed position of the particles if they do not vibrate

Trough

The lowest point of a wave

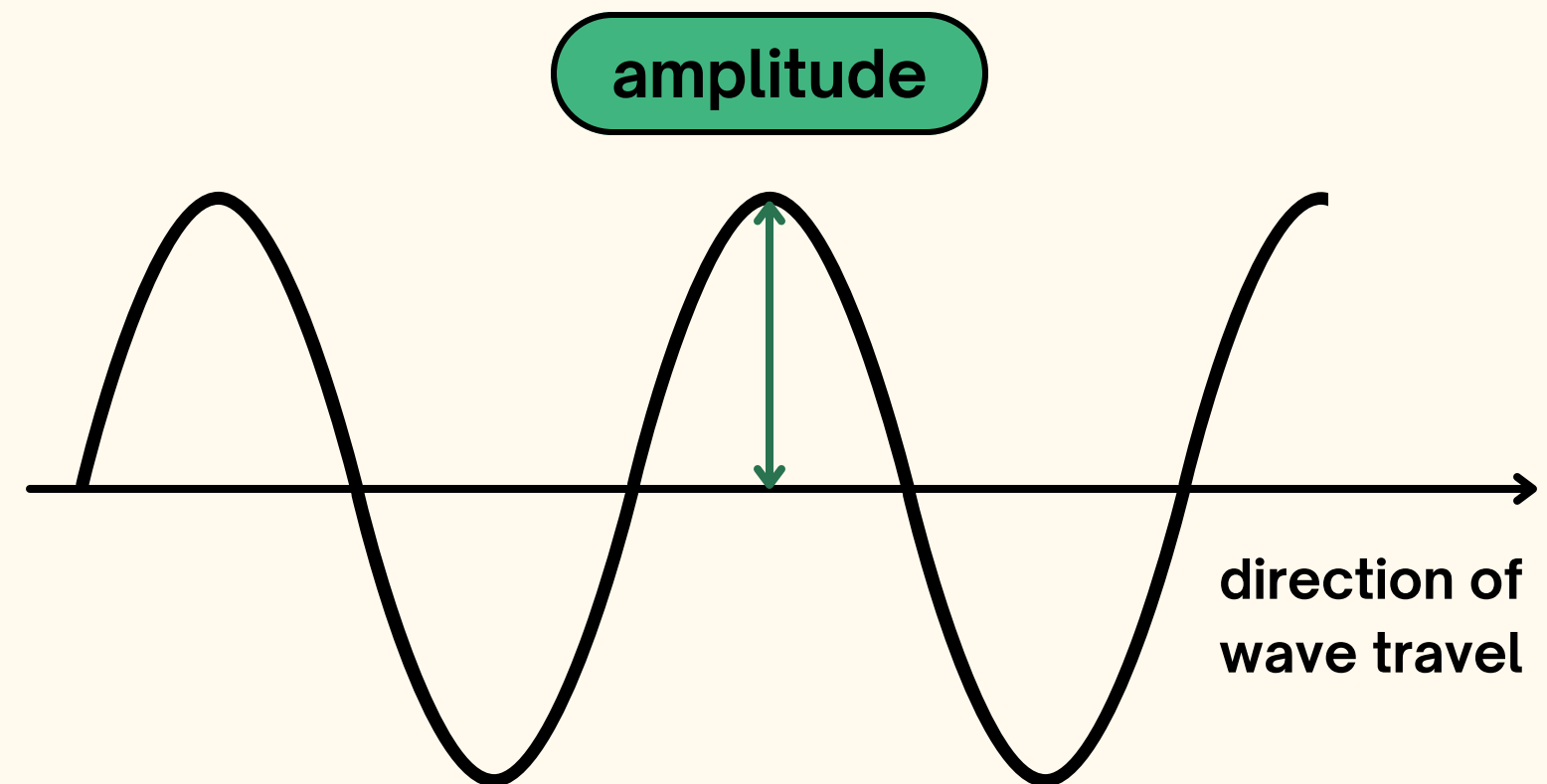


PARTS OF A TRANSVERSE WAVE:

It is the maximum disturbance from the crest of the wave from its rest position. It represents the **wave's height**.

The unit for amplitude (A) is **metres (m)**.

Amplitude

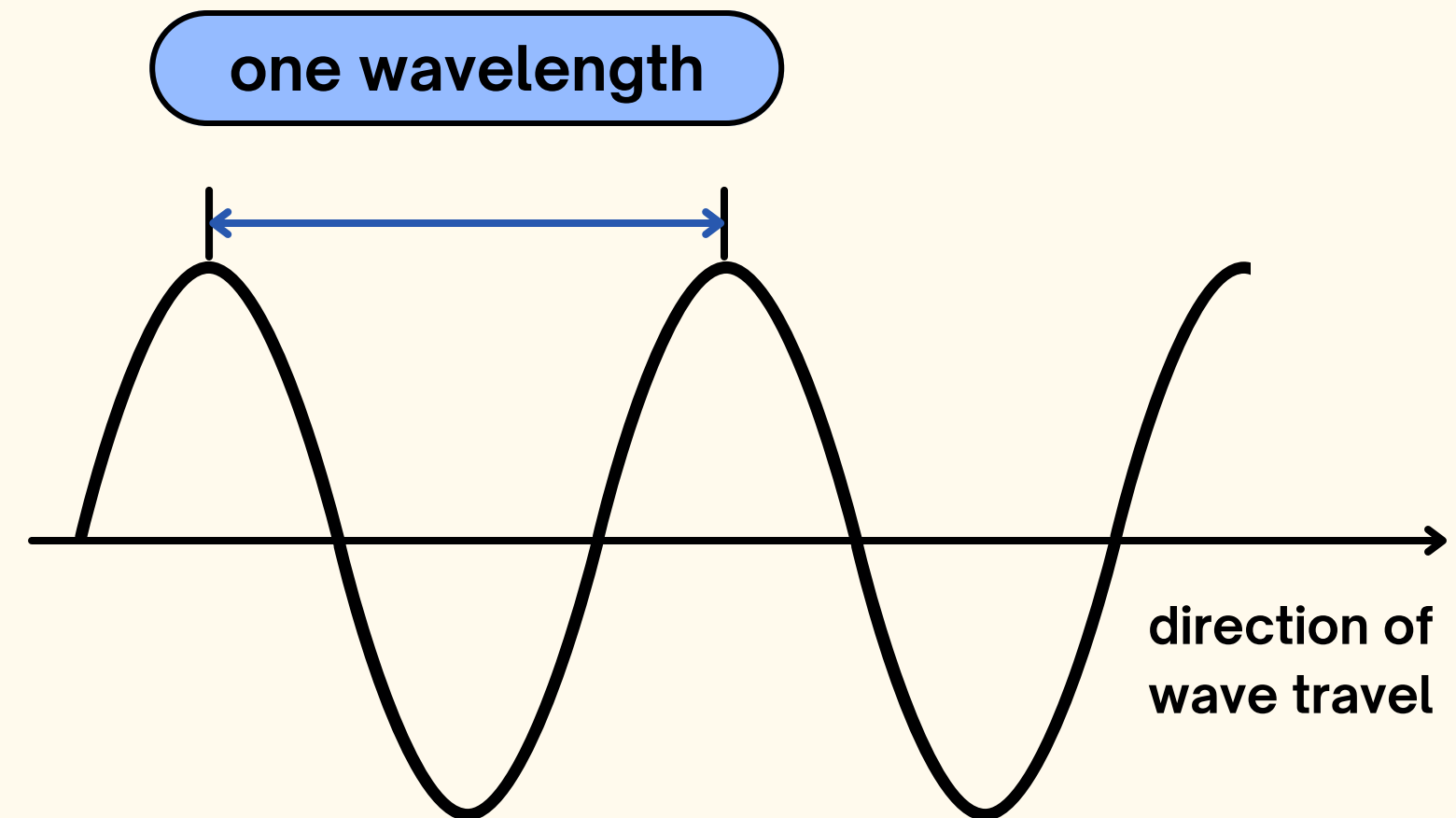


PARTS OF A TRANSVERSE WAVE:

It is a **set distance that represents the full cycle of a wave**. It can be measured from crest to crest, from trough to trough, or any two consecutive points in the wave cycle.

The unit for wavelength (λ , Greek letter lambda) is **meters (m)**.

Wavelength



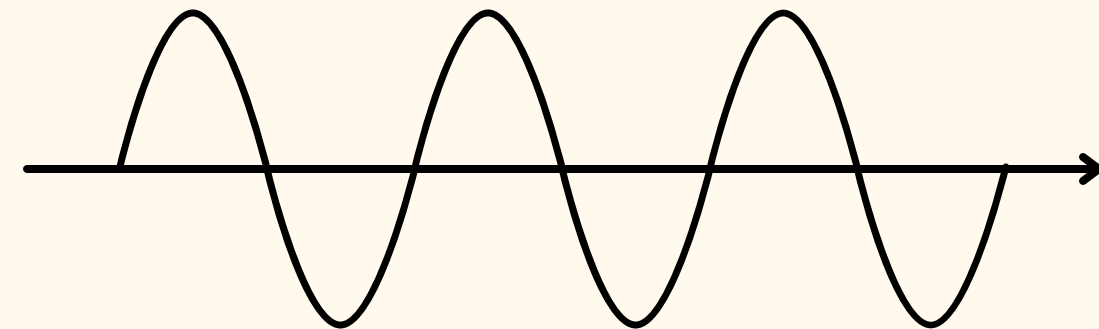
PARTS OF A TRANSVERSE WAVE:

It is the **number of waves passing a specific point per second**.

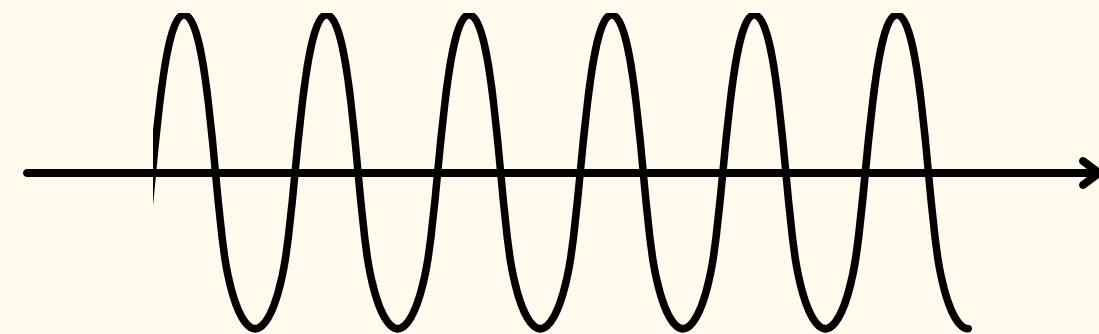
The unit of frequency is **hertz (Hz)**, which is equal to 1 cycle per second (1/s).

Frequency

low frequency



high frequency

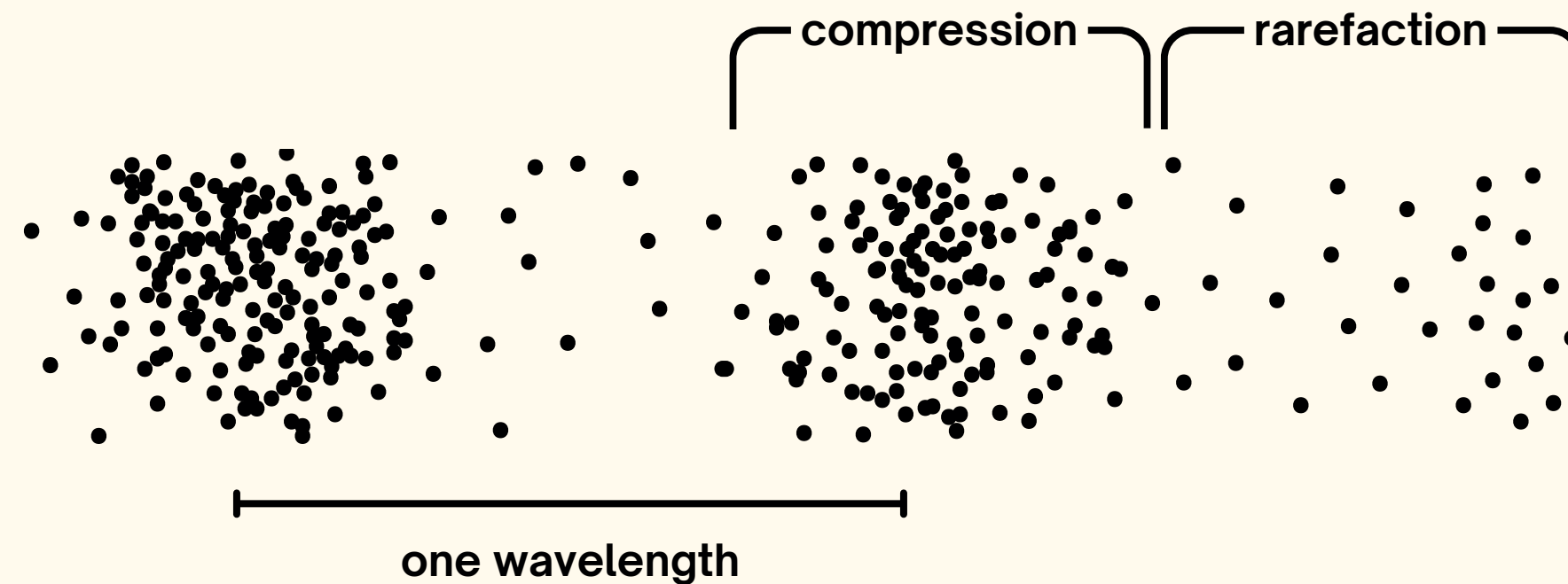


Parts of a Longitudinal Wave

Longitudinal waves also exhibit wavelength and amplitude.

Amplitude - the closer the particles are in the area of compression, the greater the amplitude

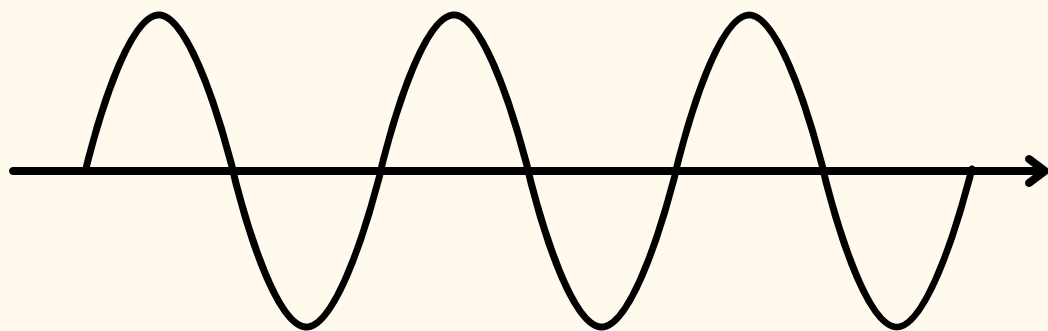
Wavelength - the distance between two consecutive points (between two compressions or two rarefactions)



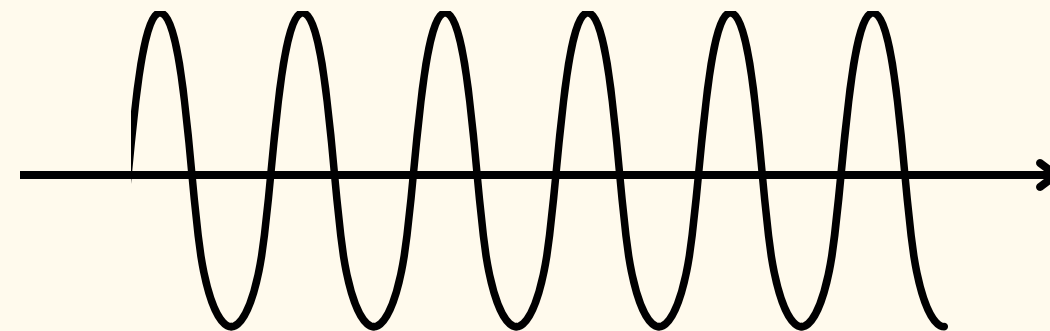
Learning Check

Compare these two waves
in terms of frequency,
wavelength, and amplitude.

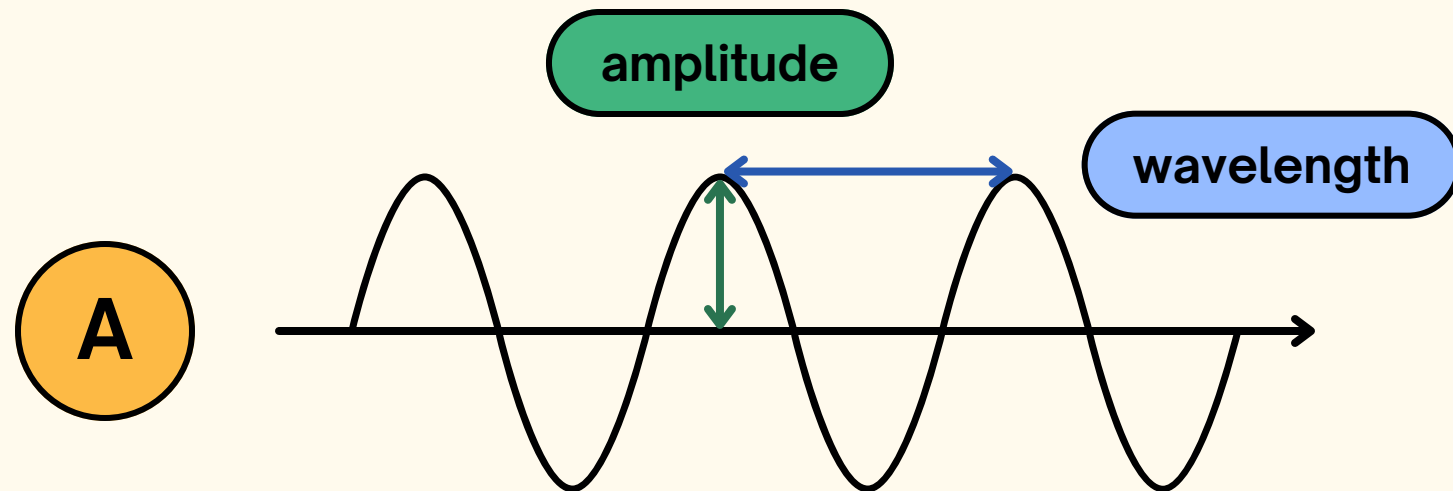
A



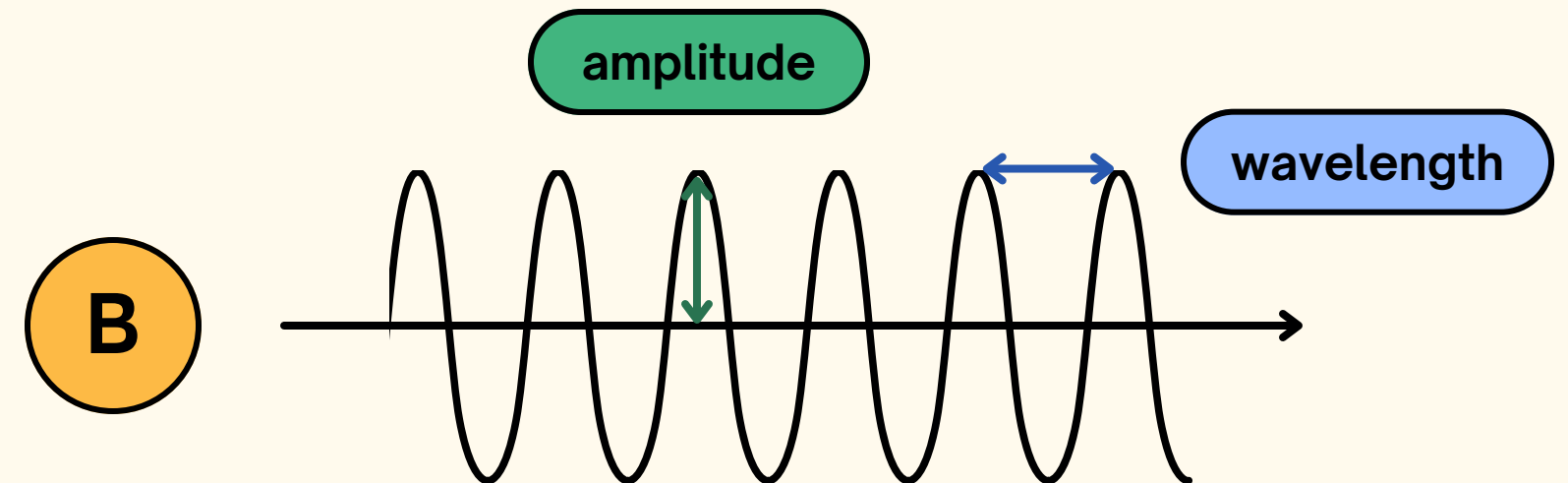
B



Learning Check



- same amplitude as B
- longer wavelength
- lower frequency

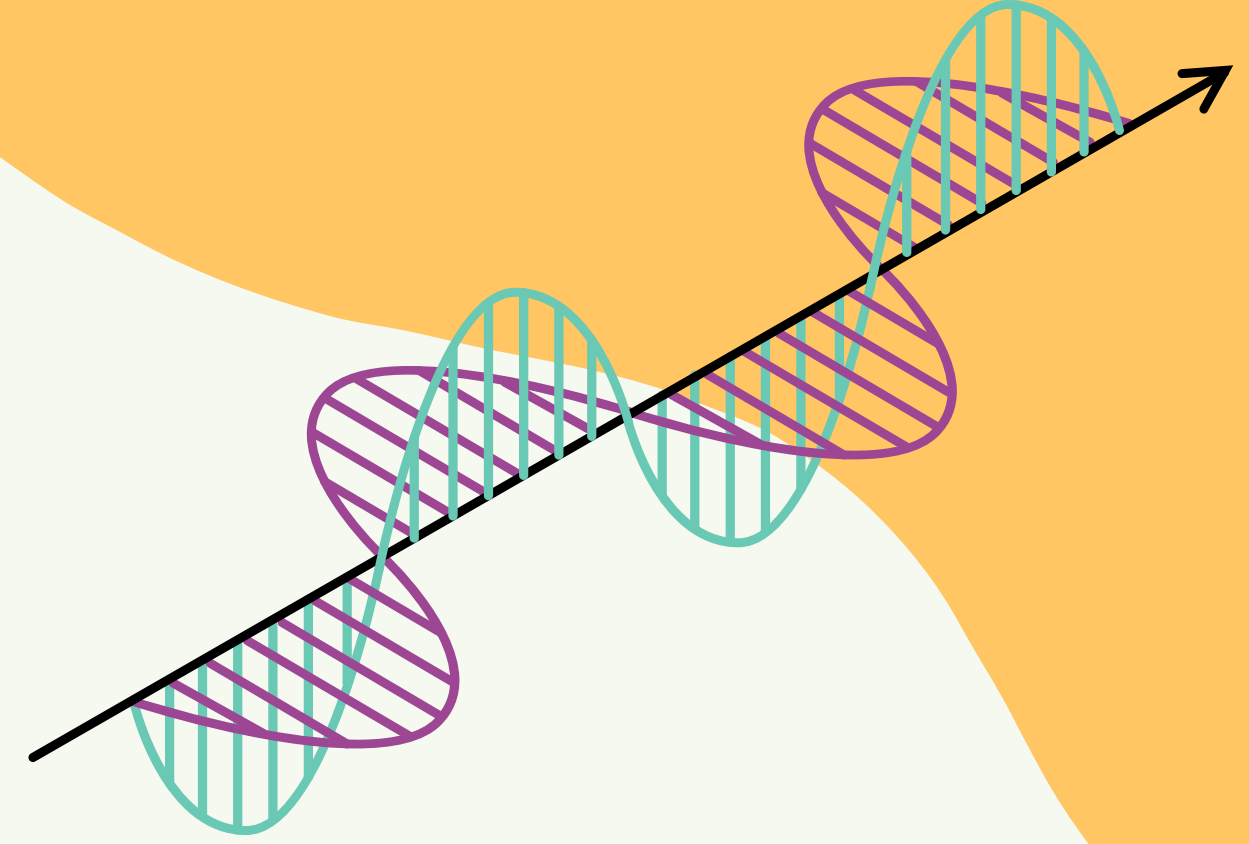


- same amplitude as A
- shorter wavelength
- higher frequency

Learning Check

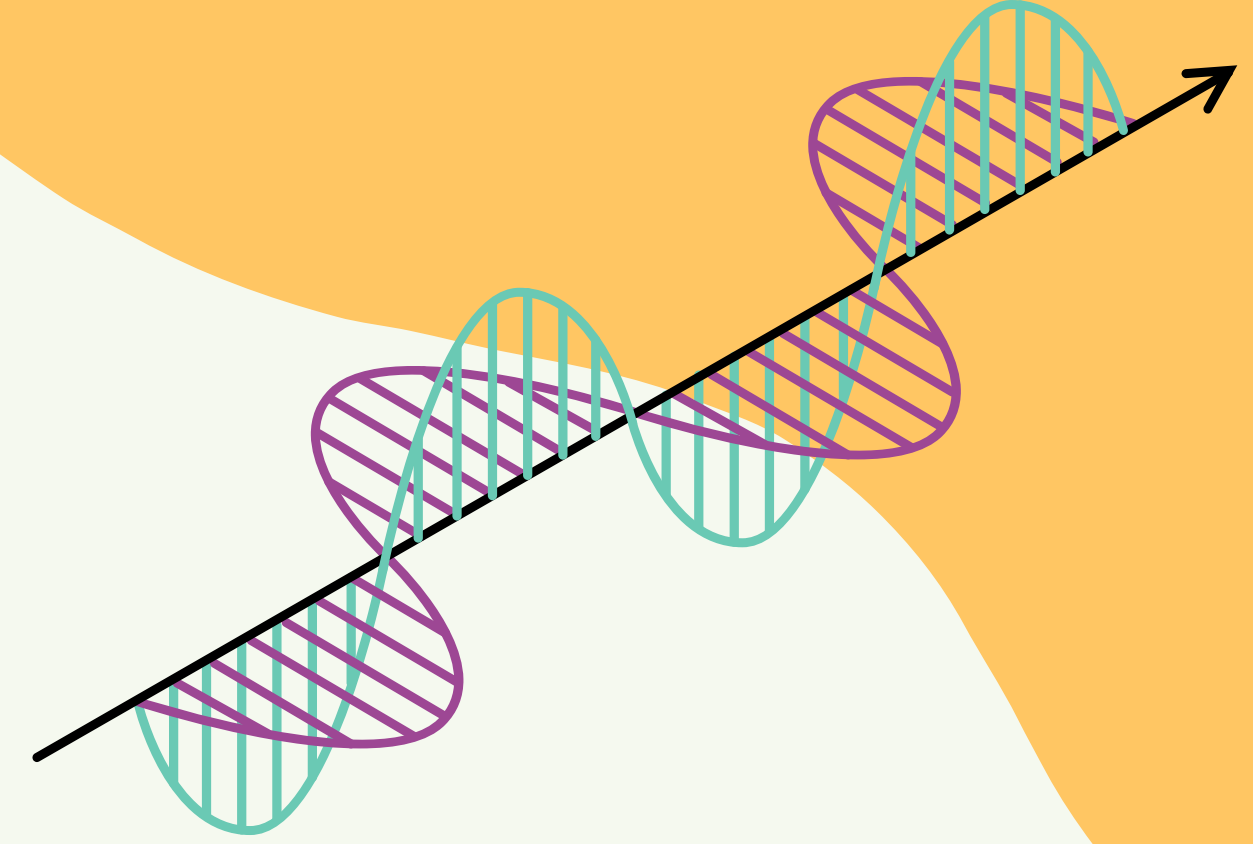
What is a wave?

- A) A disturbance that carries energy through matter or empty space**
- B) A type of particle found in the ocean**
- C) A measurement of temperature**



Learning Check

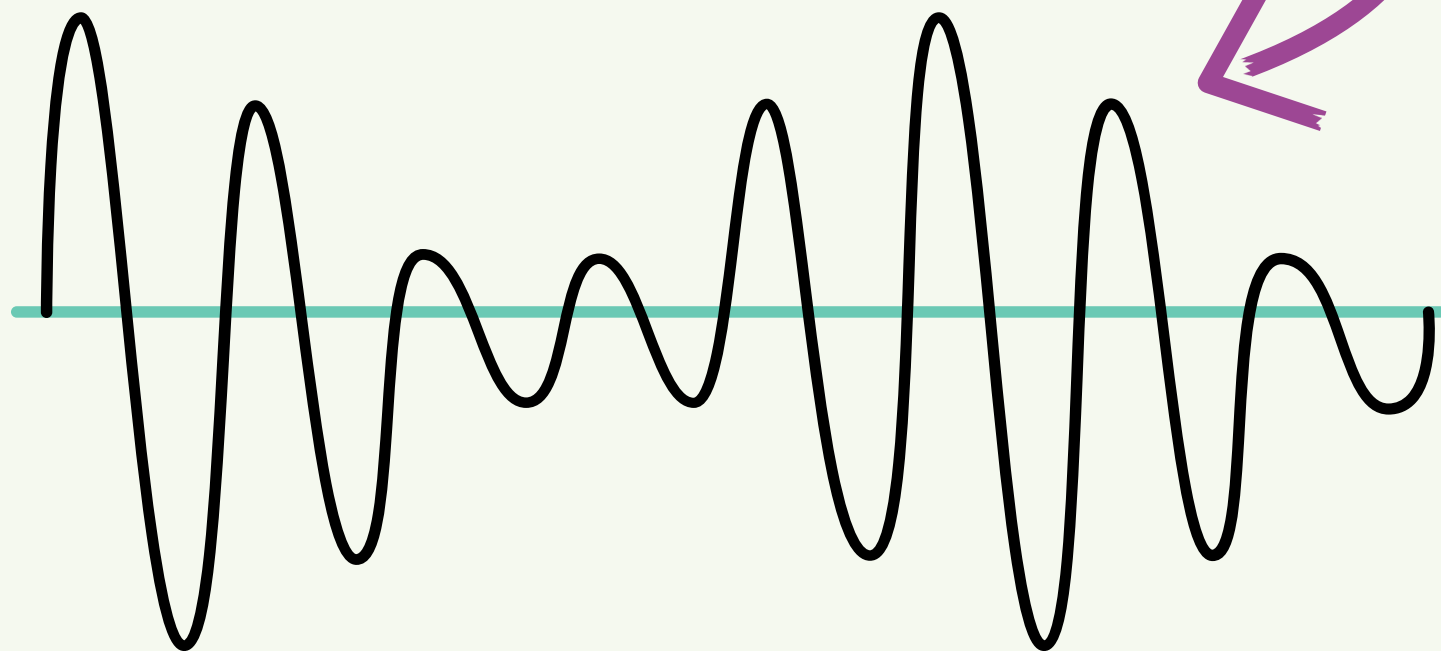
What is a wave?



- A) A disturbance that carries energy through matter or empty space**
- B) A type of particle found in the ocean**
- C) A measurement of temperature**

Learning Check

What is the
highest point of
a wave called?



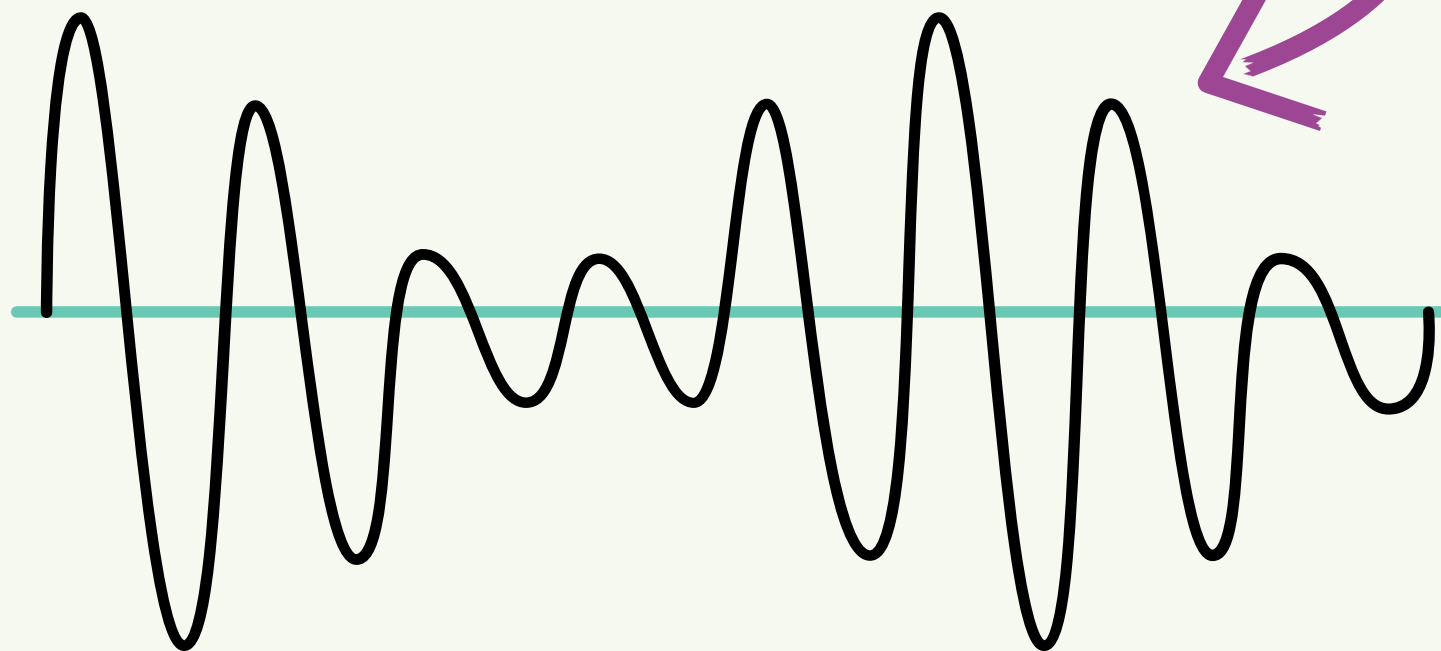
A) Trough

B) Crest

C) Amplitude

Learning Check

What is the highest point of a wave called?



A) Trough

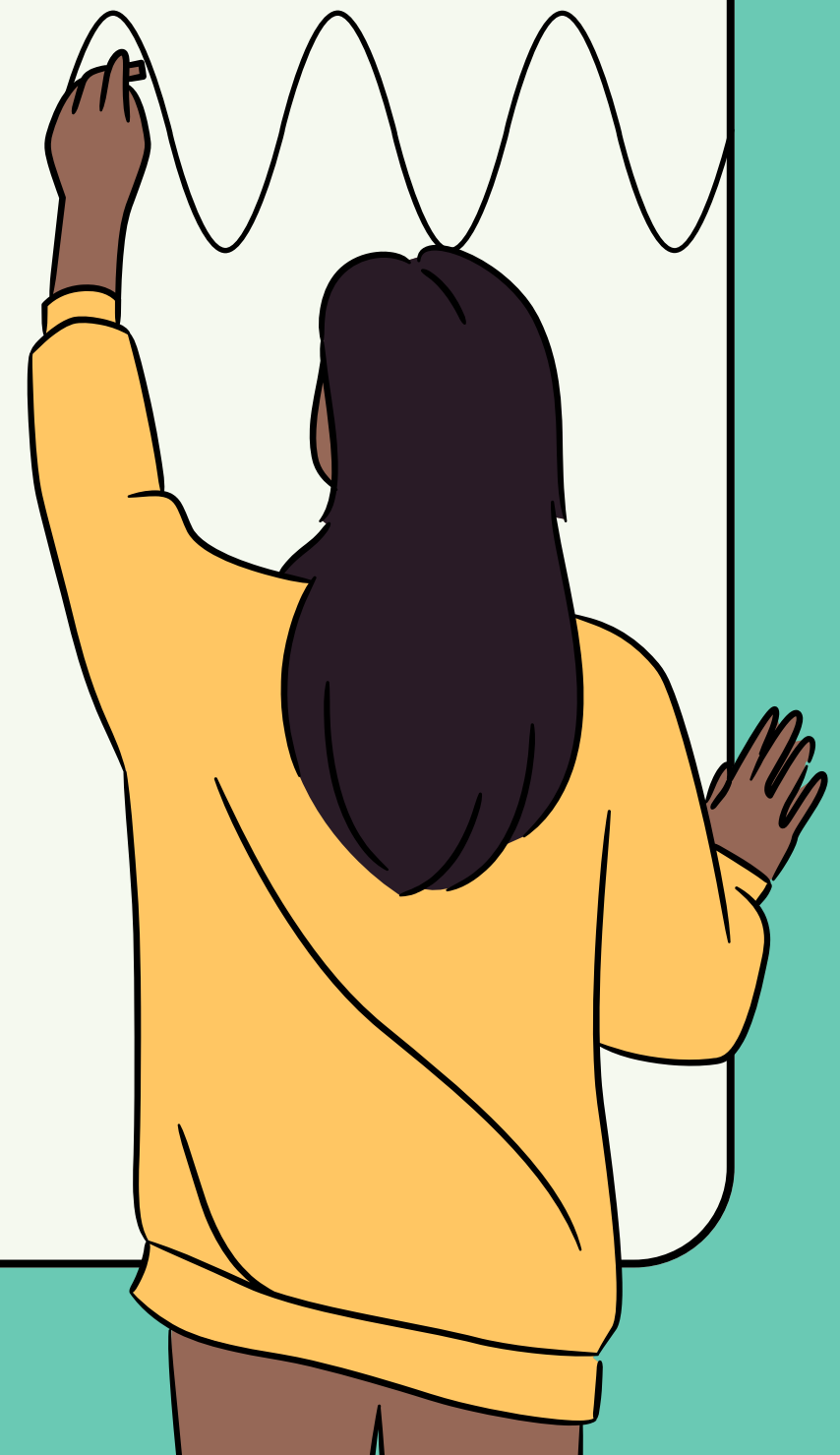
B) Crest

C) Amplitude

Learning Check

Which of the following statements is true about wavelength?

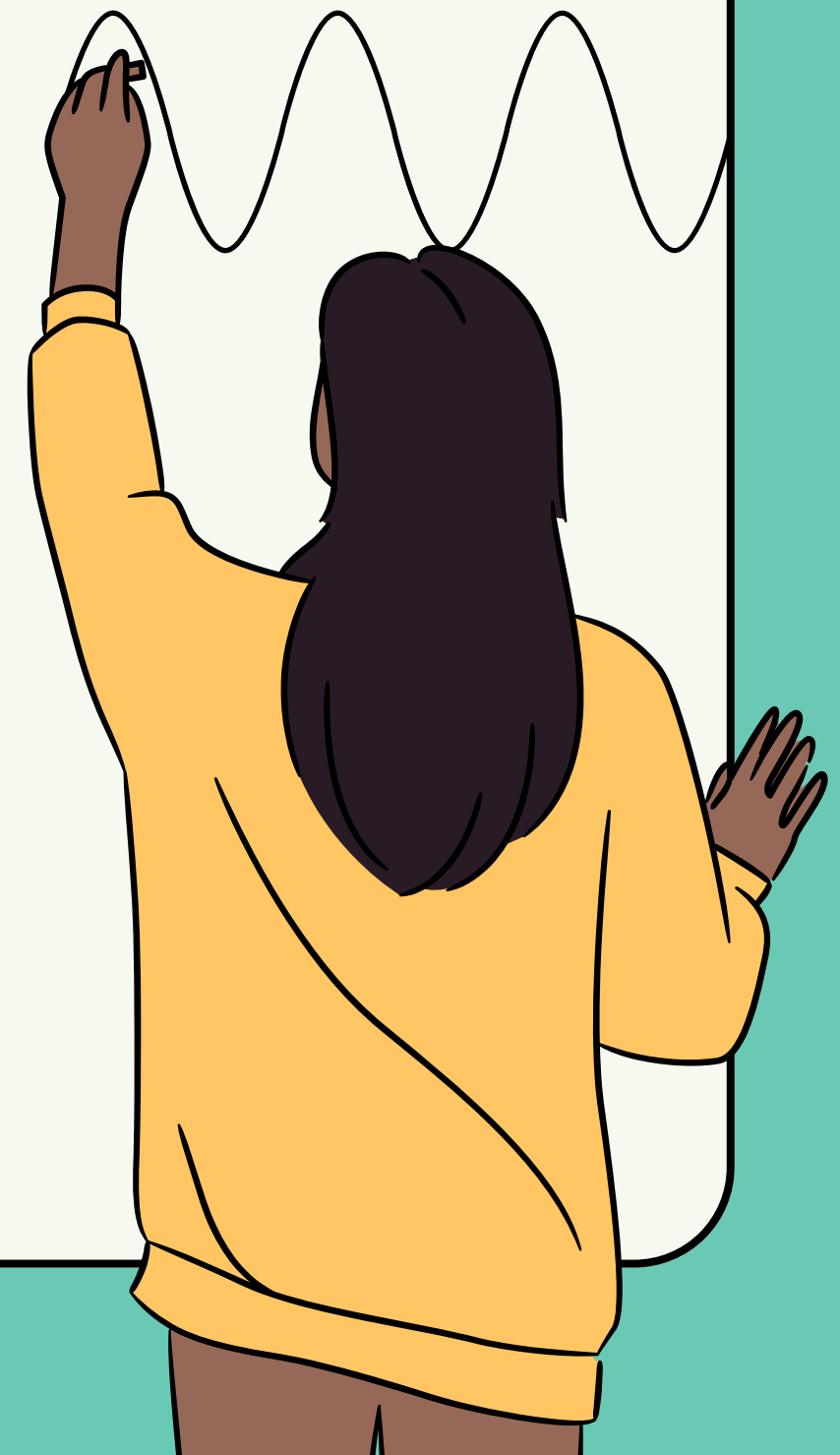
- A) It is the distance between two adjacent crests or troughs of a wave
- B) It is the height of a wave from the resting position to the crest
- C) It measures the loudness of a sound wave



Learning Check

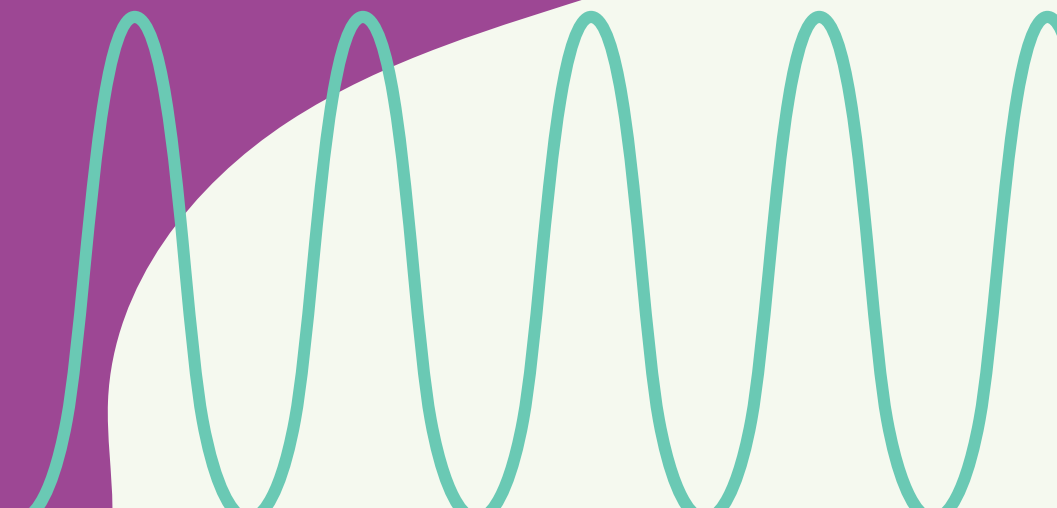
Which of the following statements is true about wavelength?

- A) It is the distance between two adjacent crests or troughs of a wave
- B) It is the height of a wave from the resting position to the crest
- C) It measures the loudness of a sound wave



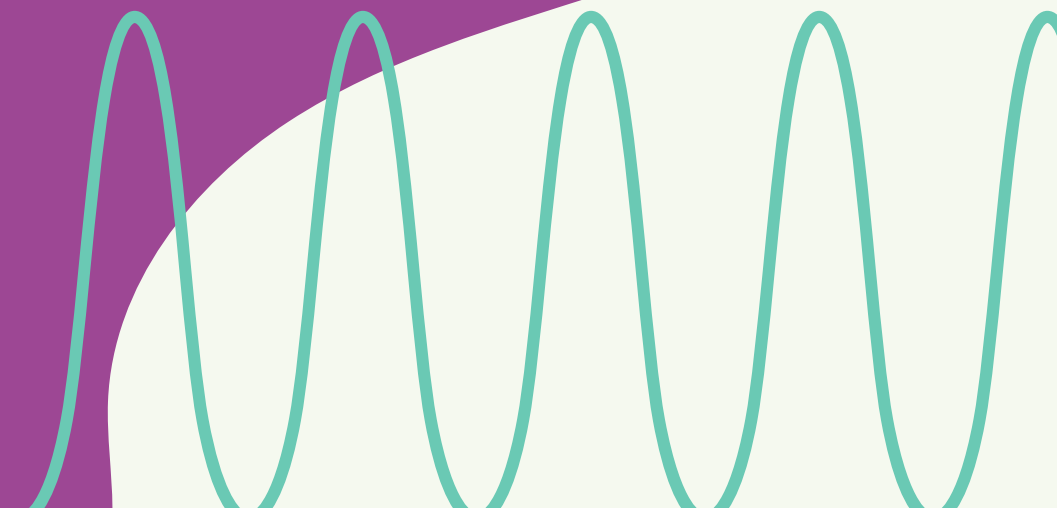
What happens to the frequency of a wave if its wavelength decreases?

- A) The frequency increases
- B) The frequency decreases
- C) The frequency remains the same

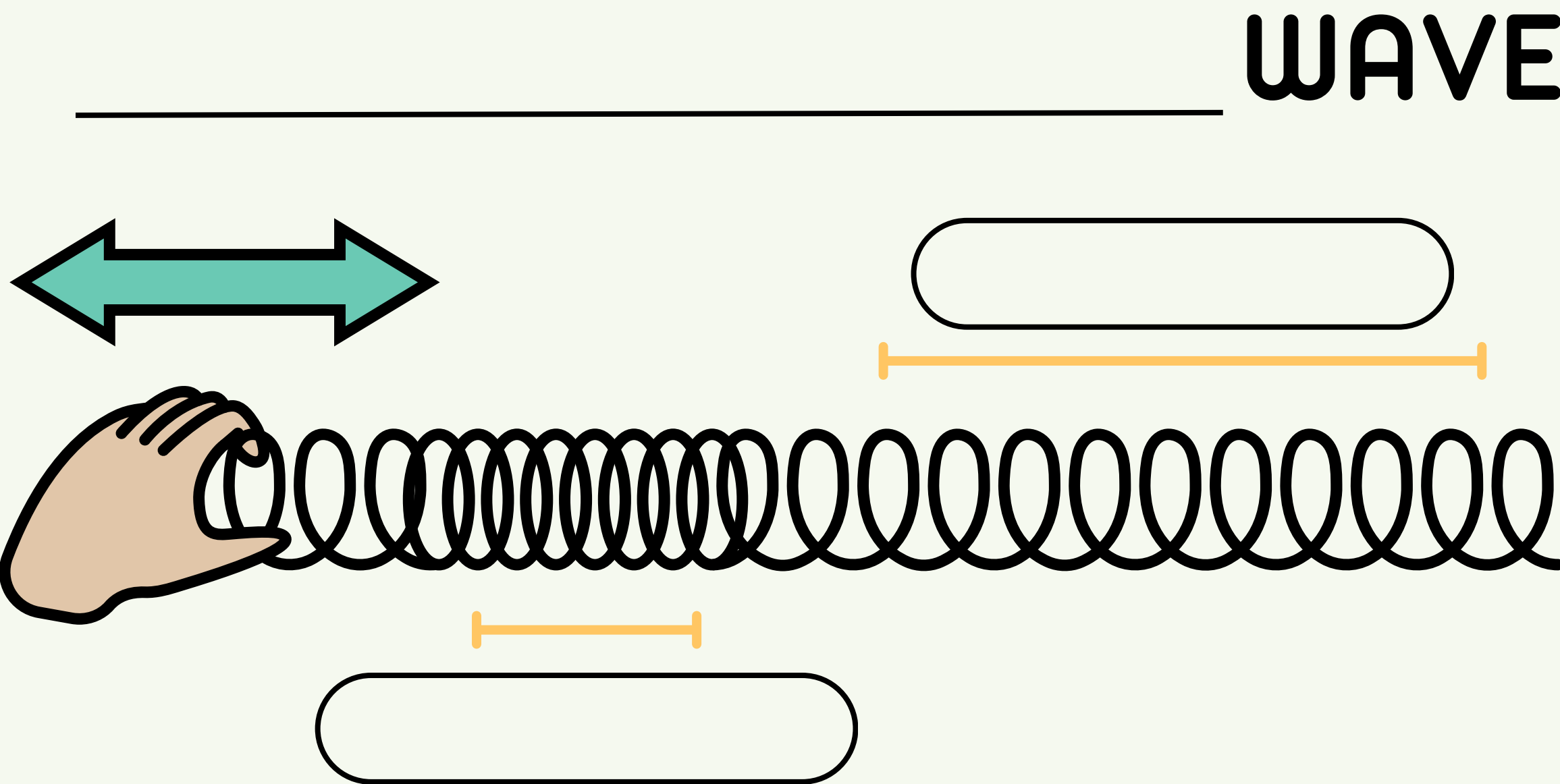


What happens to the frequency of a wave if its wavelength decreases?

- A) The frequency increases
- B) The frequency decreases
- C) The frequency remains the same

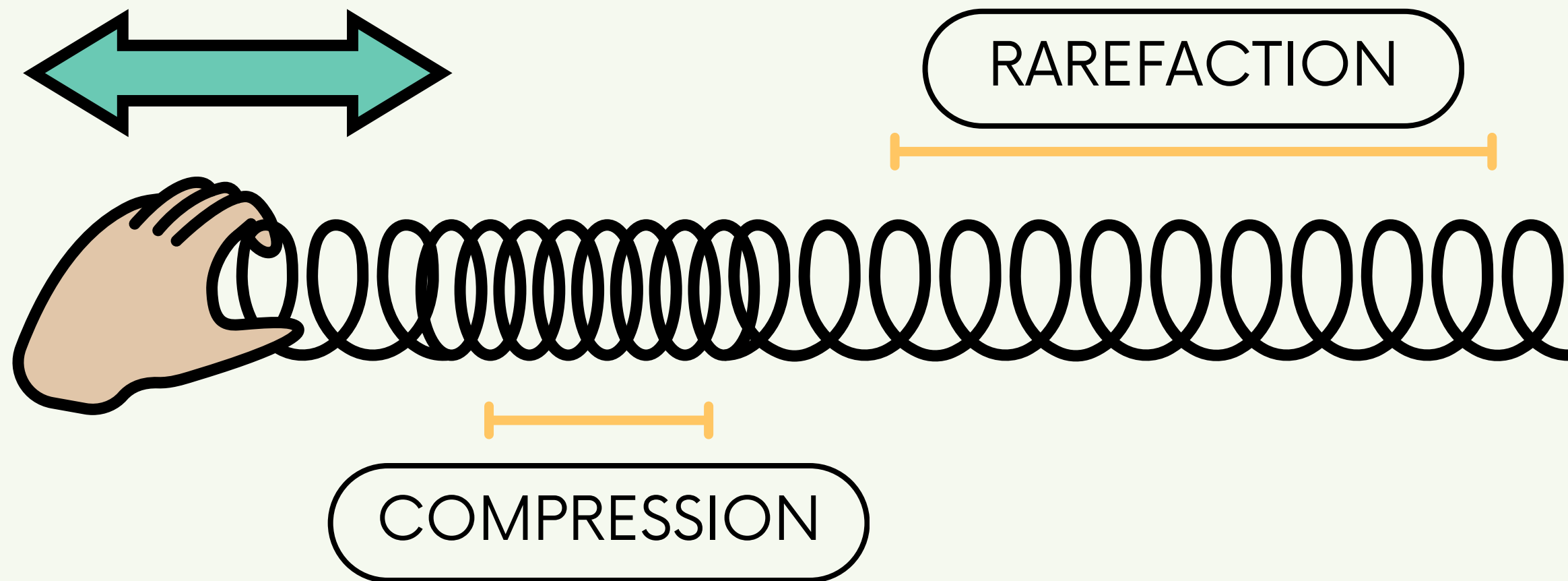


Name and label the type of wave illustrated below.

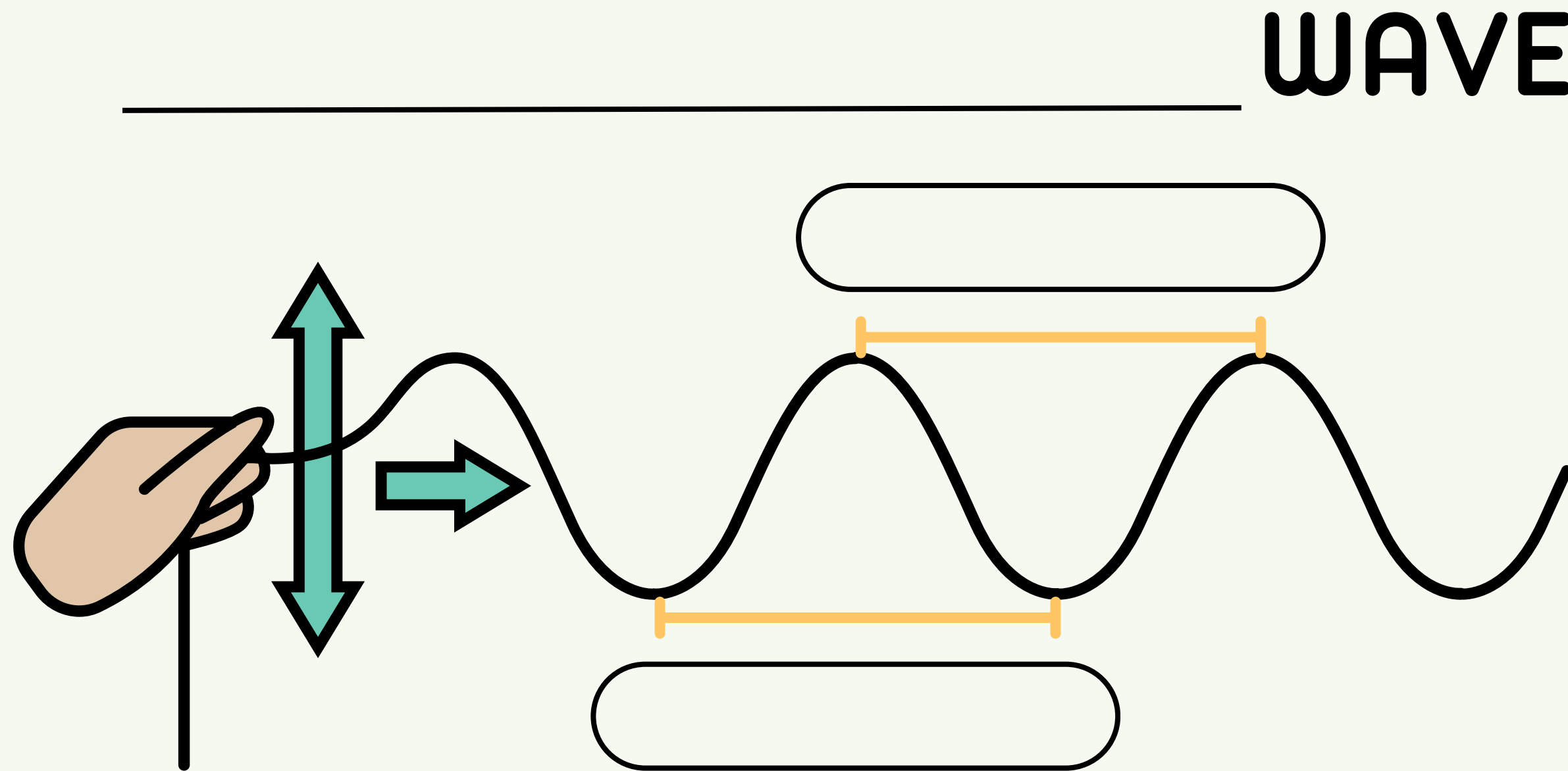


Name and label the type of wave illustrated below.

LONGITUDINAL WAVE

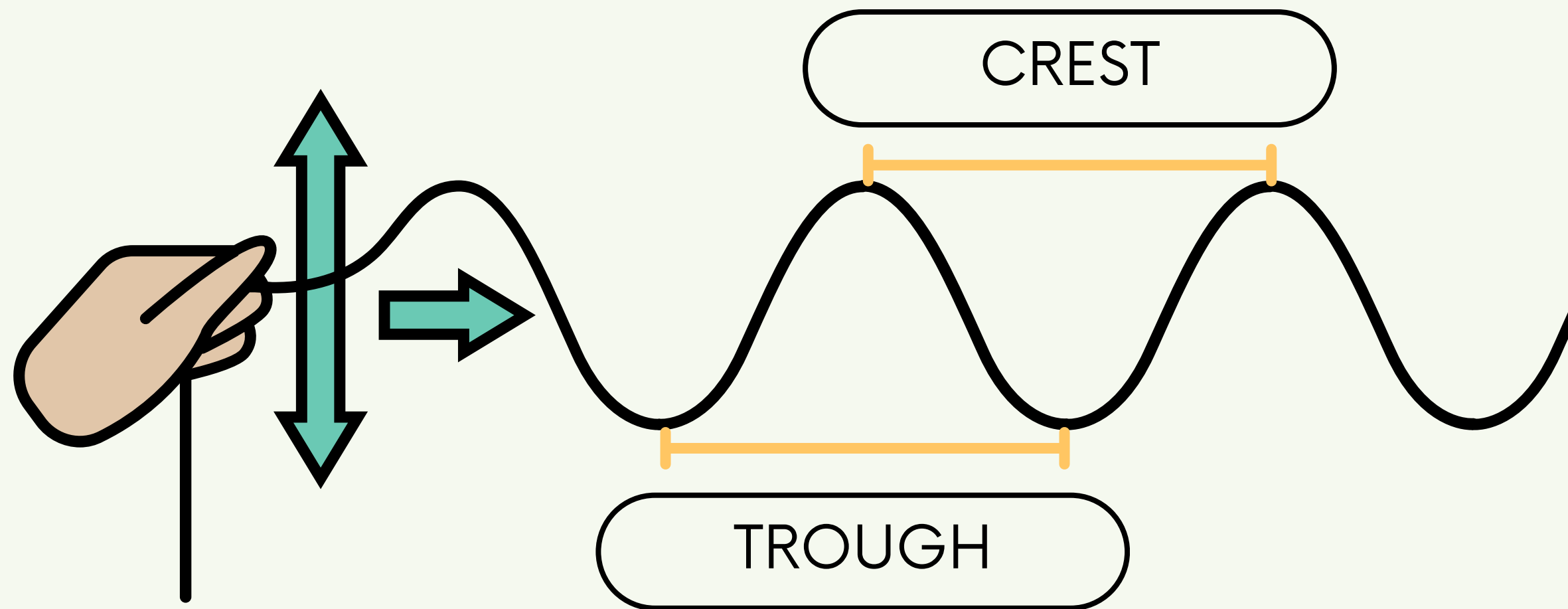


Name and label the type of wave illustrated below.



Name and label the type of wave illustrated below.

TRANSVERSE WAVE





Types of Waves

**according to
Need for a
Medium**

Mechanical Waves

Mechanical waves, e.g., sound waves, **require a medium (particles) to travel through**. Sound and water waves are examples of mechanical waves.

Electromagnetic Waves

They do not require a medium to travel through and **can travel through a vacuum**. Visible light, radio waves, and gamma rays are some examples of electromagnetic waves.

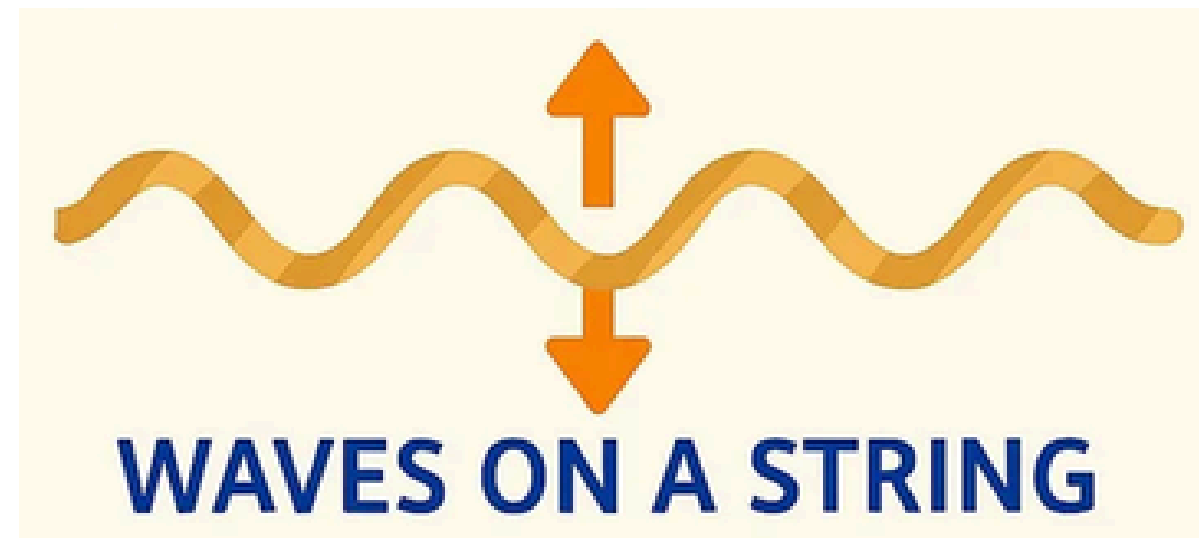
MECHANICAL WAVES

A mechanical wave is a disturbance that moves through matter, carrying energy from one place to another. It **needs a material medium**—like a solid, liquid, or gas—to travel because the wave moves by making nearby particles bump into each other.

Mechanical waves can be a transverse, longitudinal, or surface waves.



Longitudinal
Wave

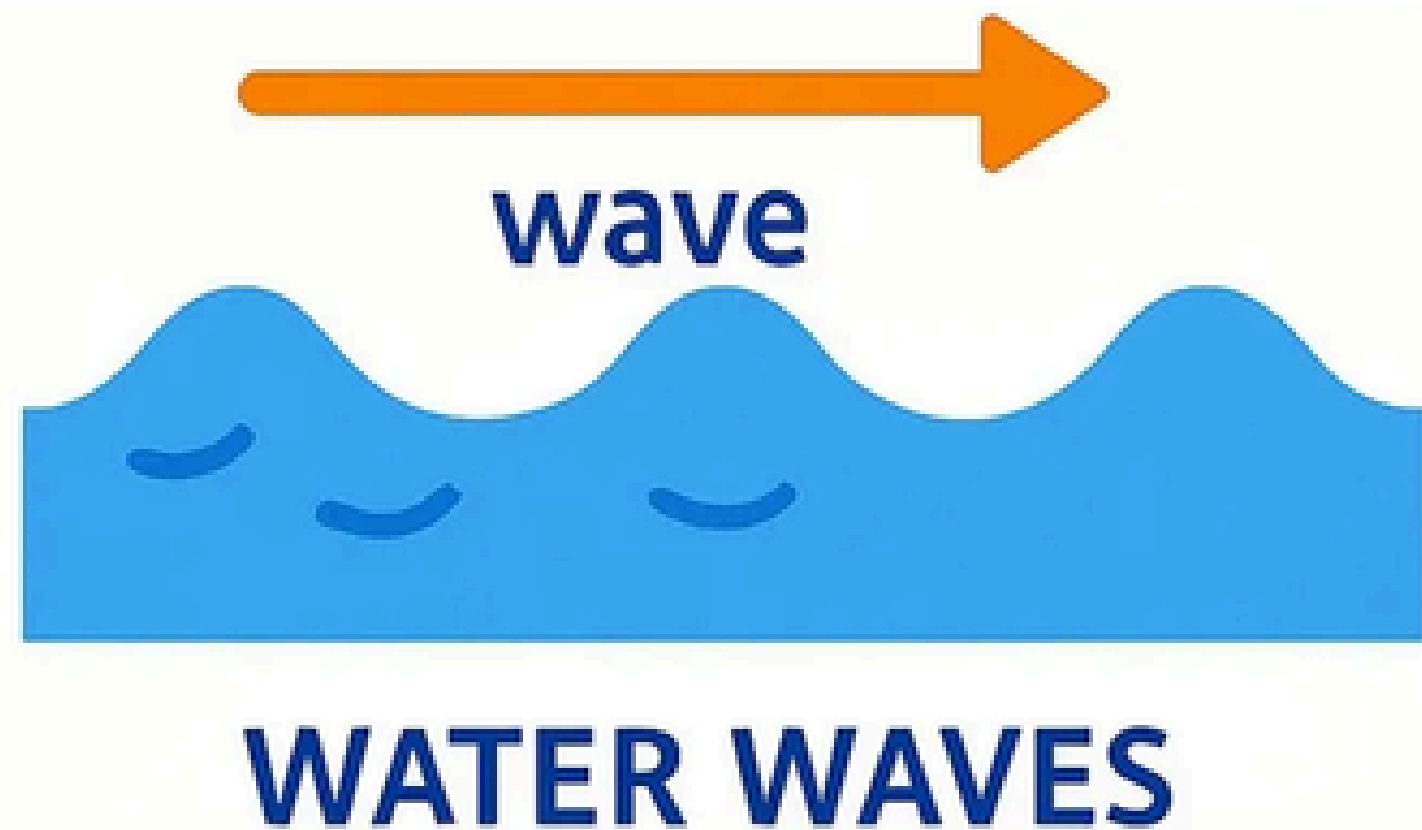


Transverse
Wave



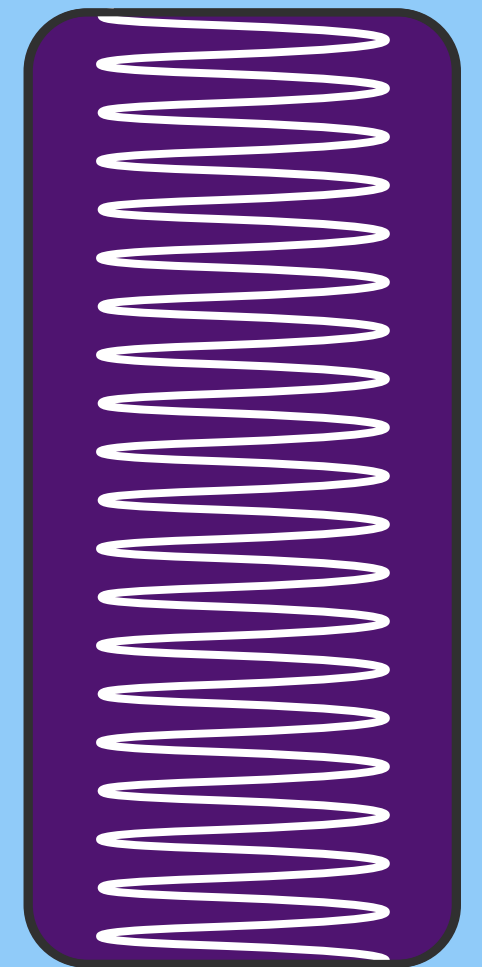
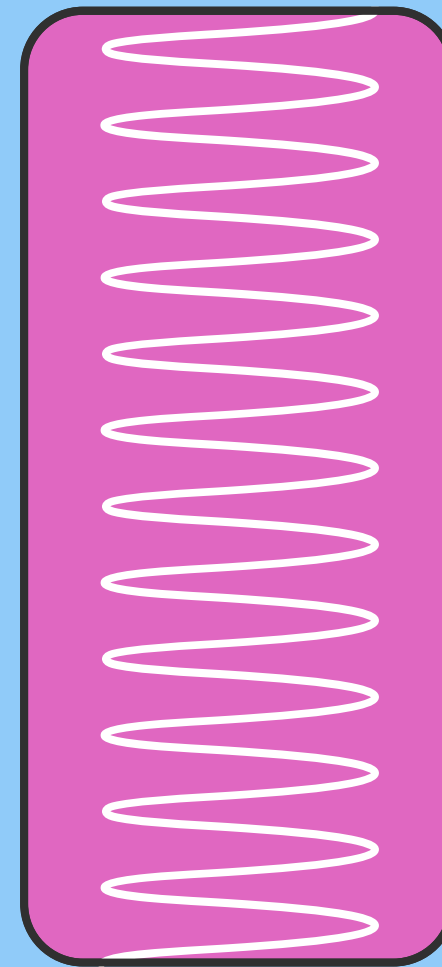
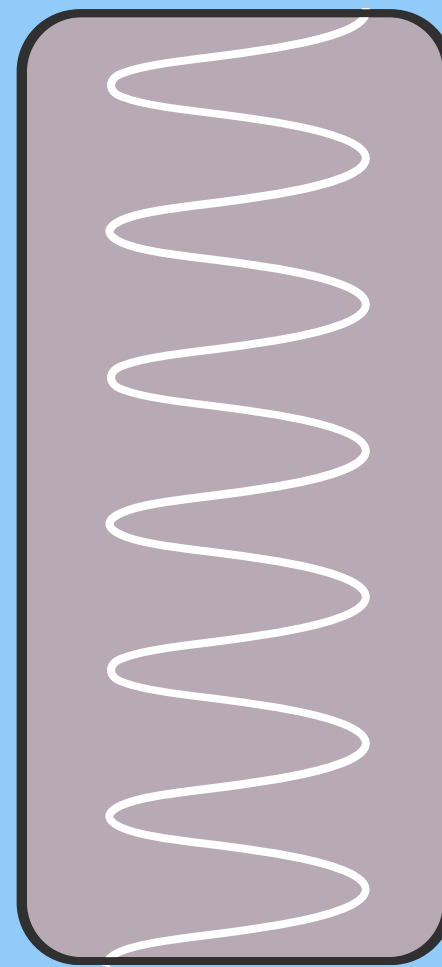
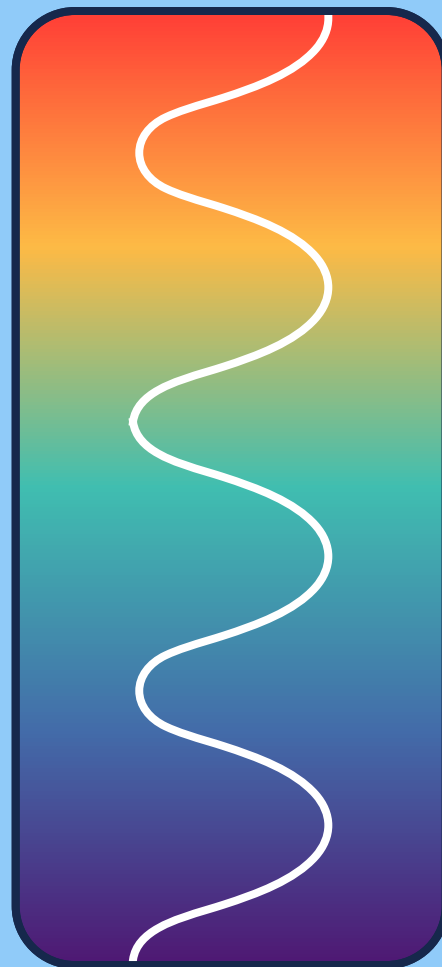
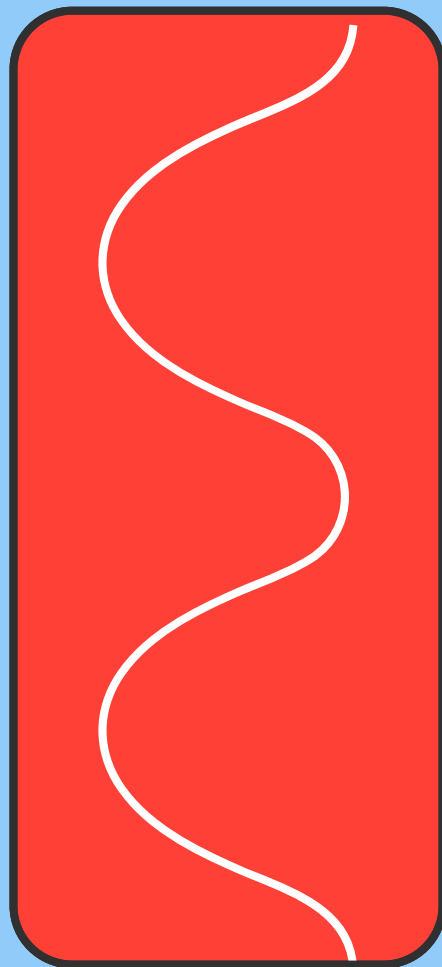
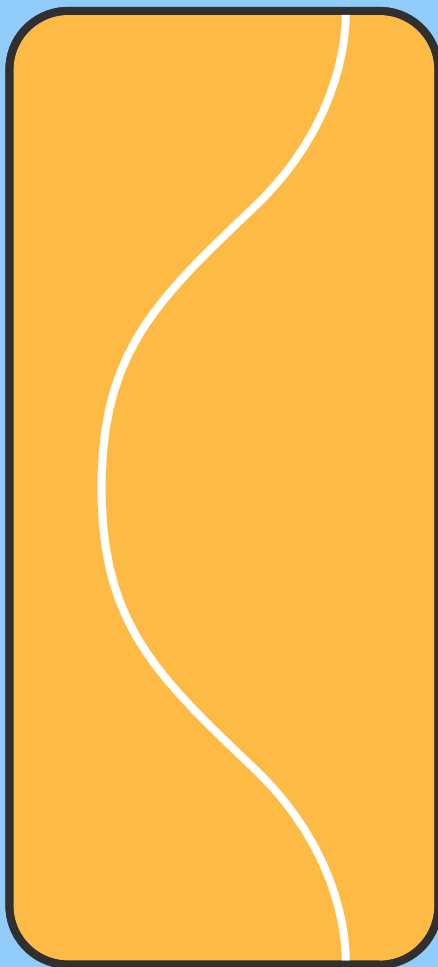
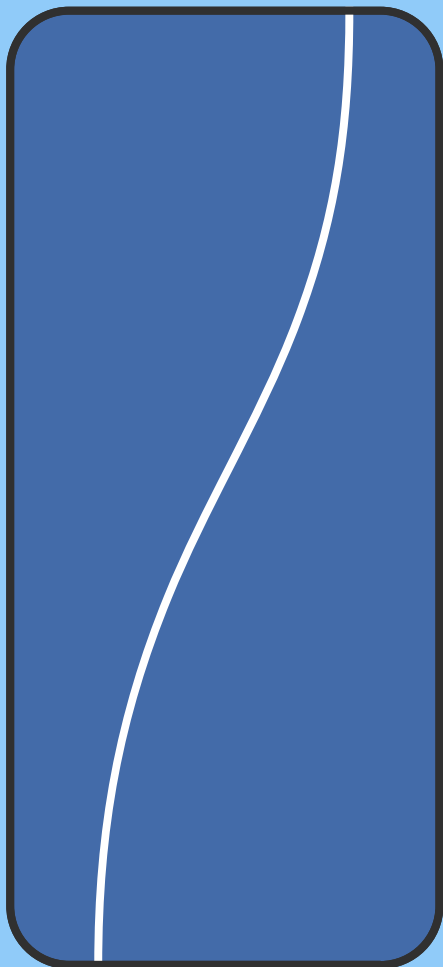
P-waves - longitudinal
S-waves - transverse

A surface wave is a wave that travels along the boundary between two different media, with particles moving in a combination of transverse and longitudinal directions.

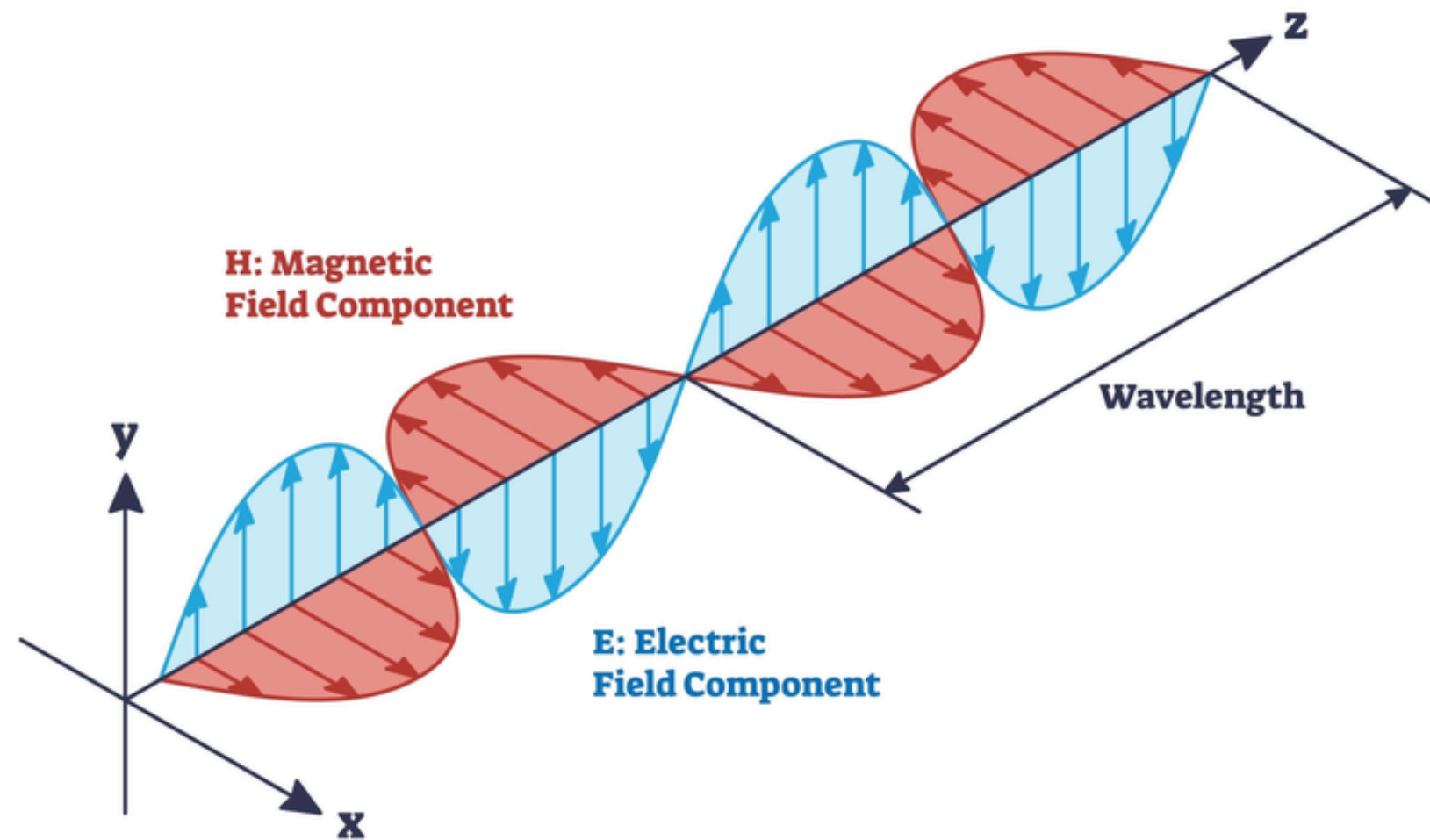


Water waves are surface waves because they travel along the boundary between air and water.

ELECTROMAGNETIC WAVES



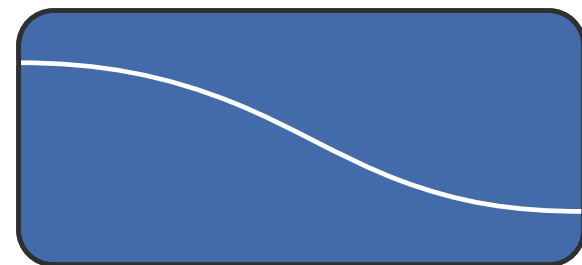
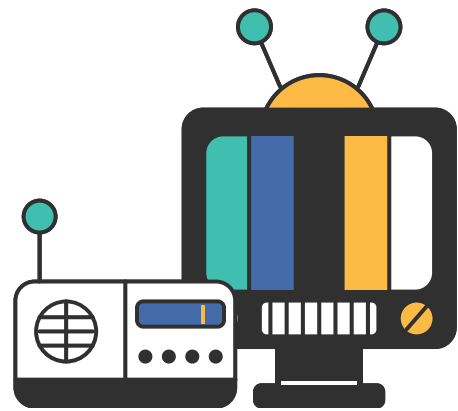
ELECTROMAGNETIC WAVES



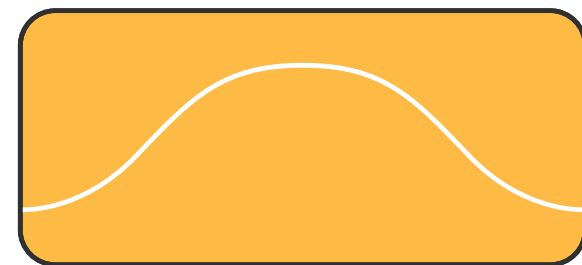
Electromagnetic waves are waves made of oscillating electric and magnetic fields that transfer energy through space.

- Do not require a medium to travel
- Can travel through a vacuum
- Are transverse waves

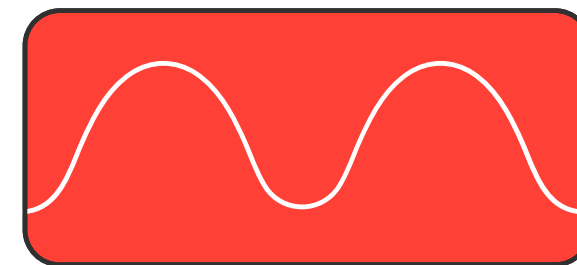
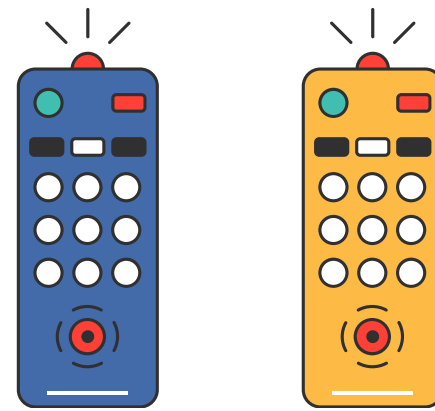
TYPES OF ELECTROMAGNETIC RADIATION



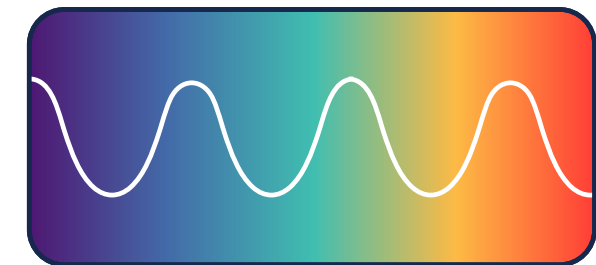
Radio waves



Microwaves

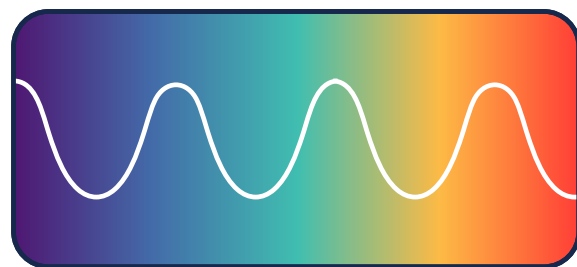
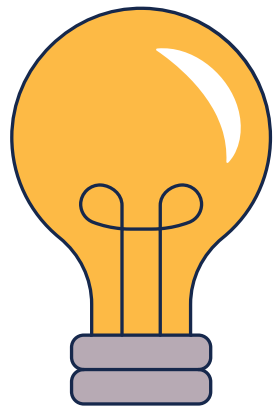


Infrared

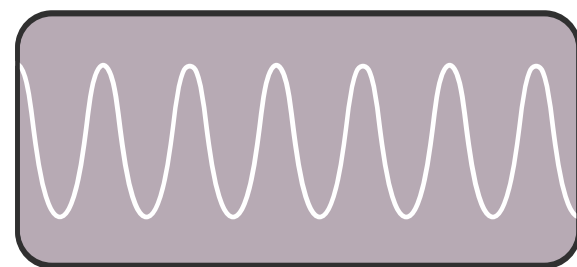
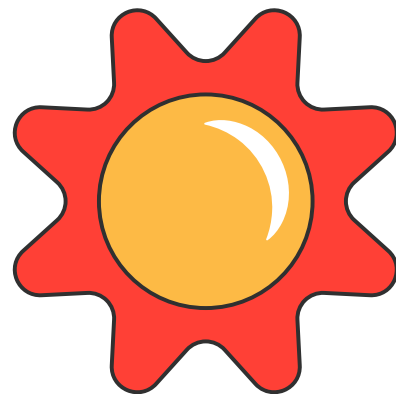


Visible Light

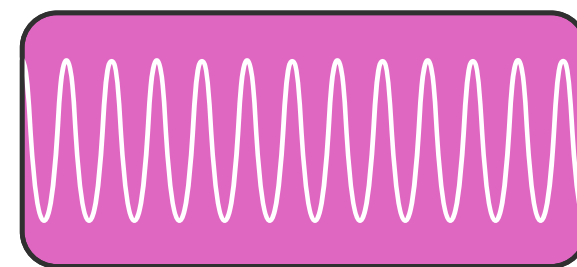
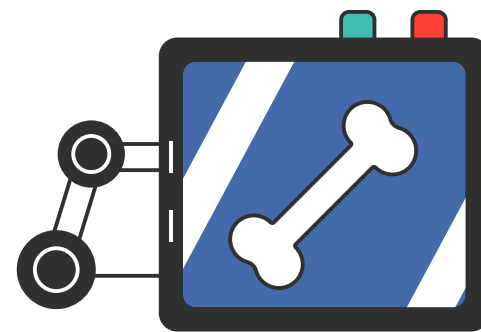
TYPES OF ELECTROMAGNETIC RADIATION



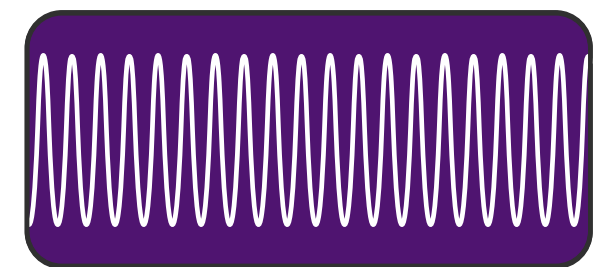
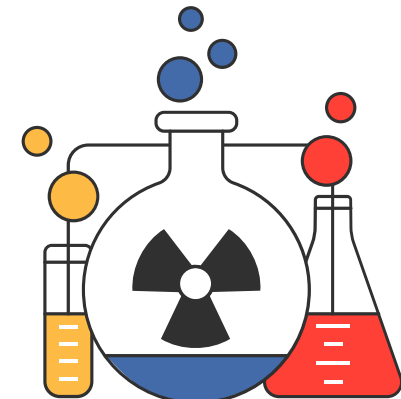
Visible Light



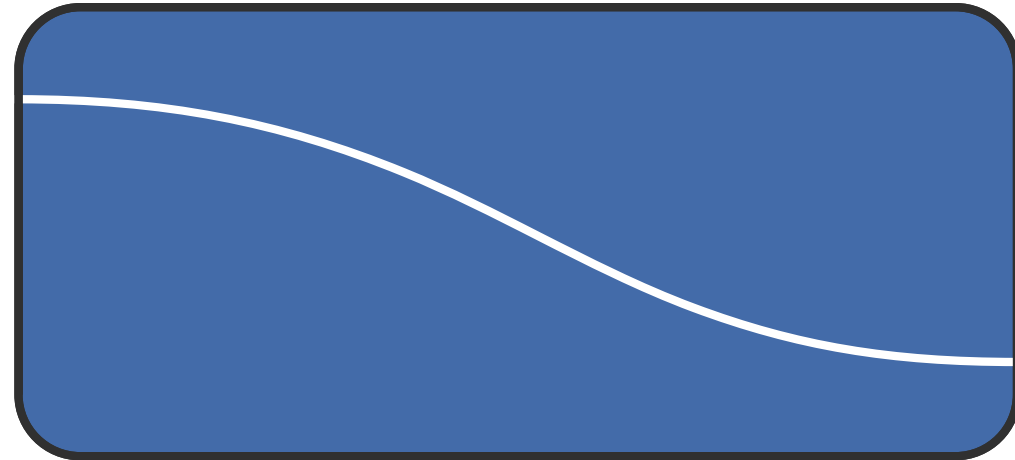
Ultraviolet



X-Radiation

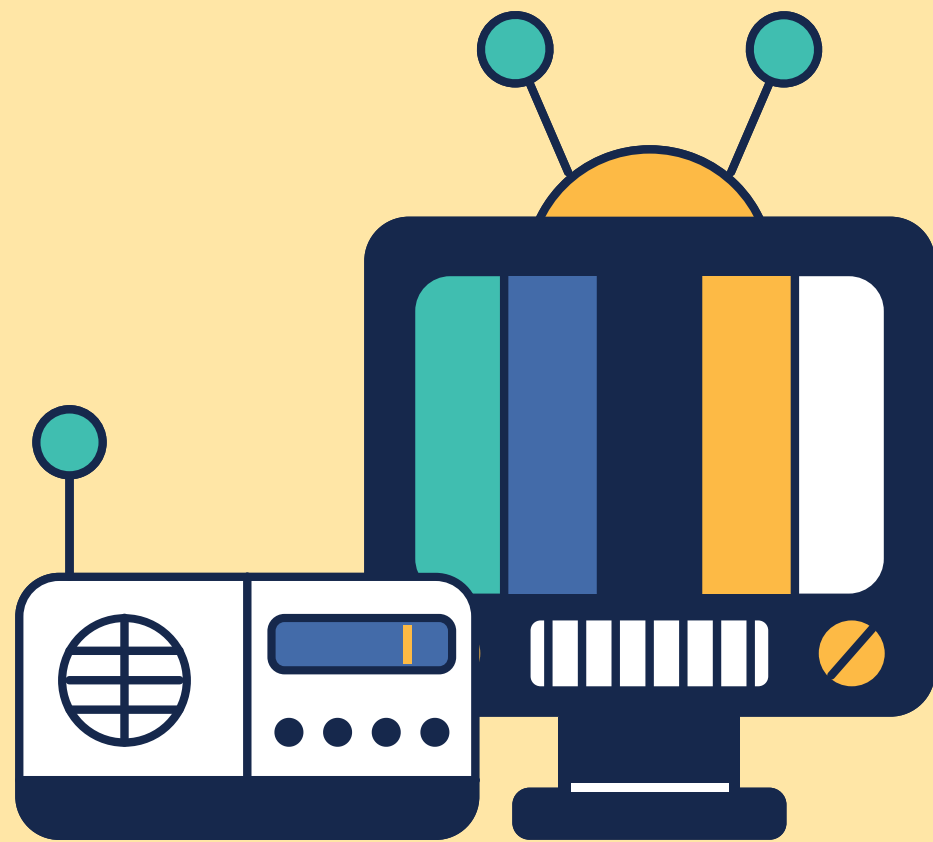


Gamma Ray



01

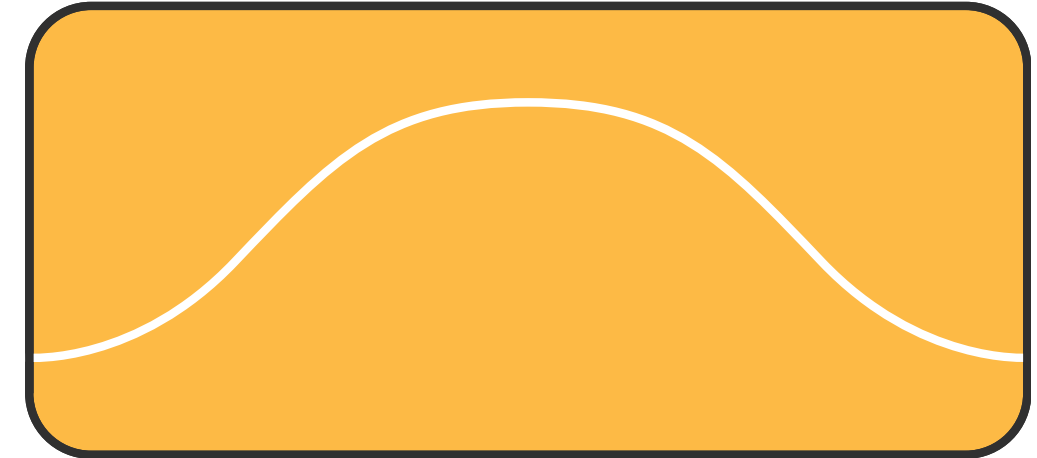
RADIO WAVES



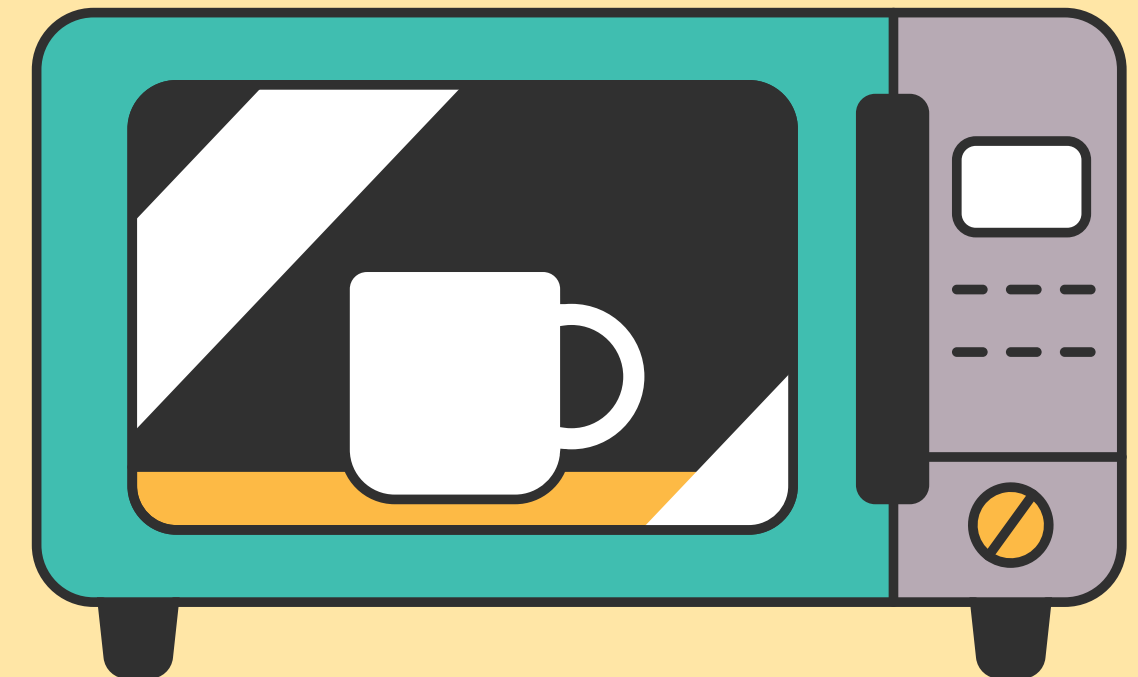
These are the waves that allow us to listen to the radio. They have long wavelengths and low frequencies.

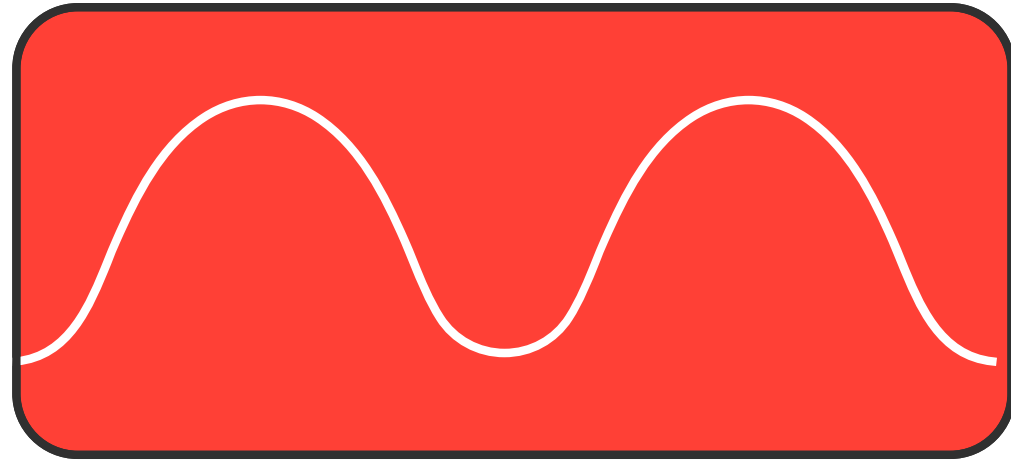
Radio waves are used for broadcasting music, news, and other information.

02 MICRO WAVES

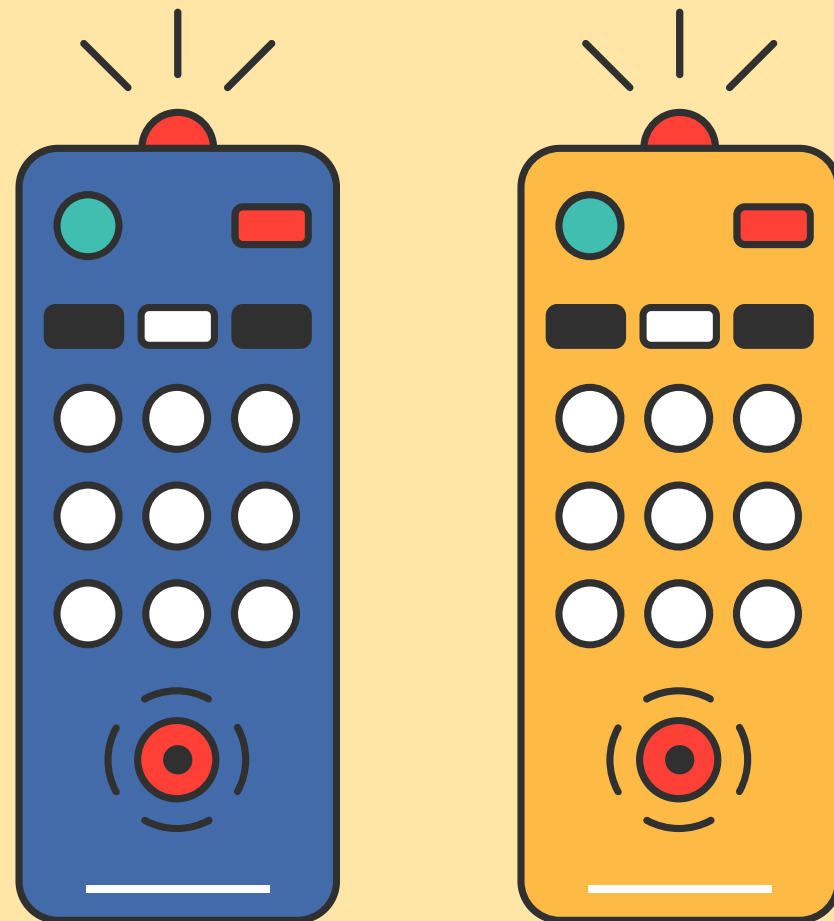


Microwaves have shorter wavelengths and higher frequencies than radio waves. They are used in microwave ovens to cook and heat food. They are also used for communication, such as cell phones and Wi-Fi.



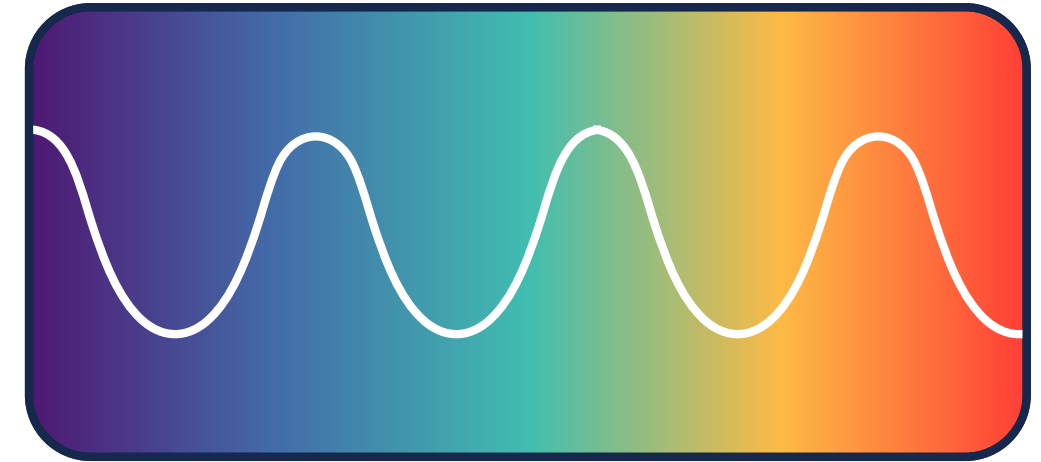


03 INFRARED RADIATION

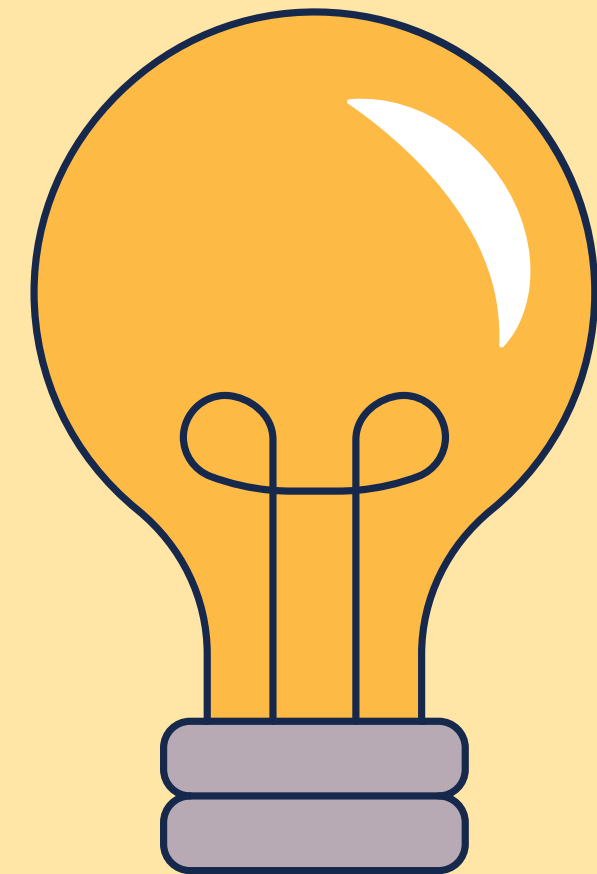


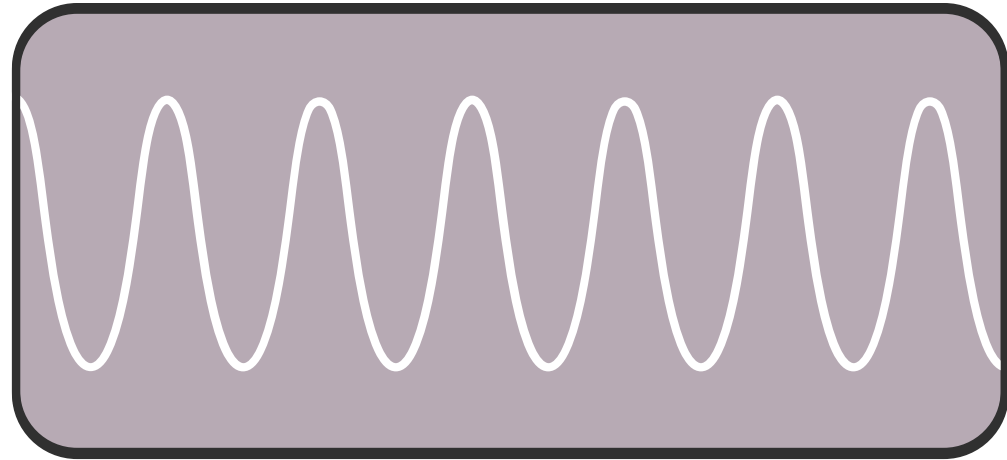
Infrared radiation is sometimes called "heat radiation" because we feel it as warmth. It has longer wavelengths than visible light. Infrared radiation is used in devices like TV remote controls and thermal cameras.

04 **VISIBLE LIGHT**

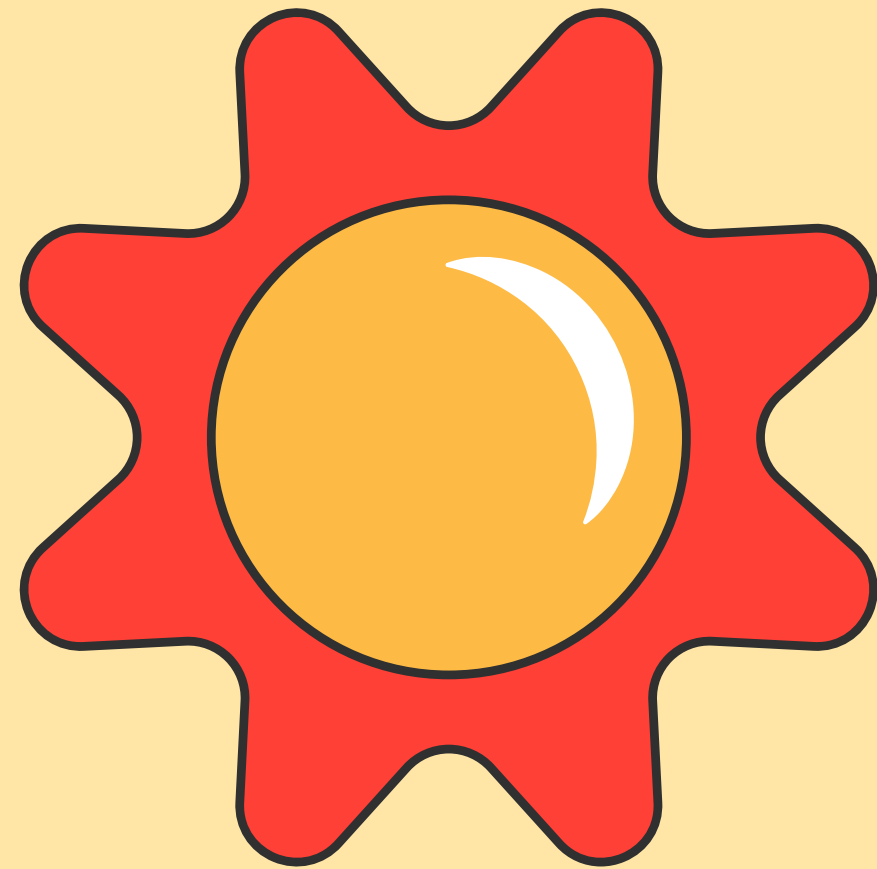


This is the type of electromagnetic radiation that allows us to see the world around us. Visible light comes in different colors, like red, orange, yellow, green, blue, indigo, and violet. Each color has a different wavelength.



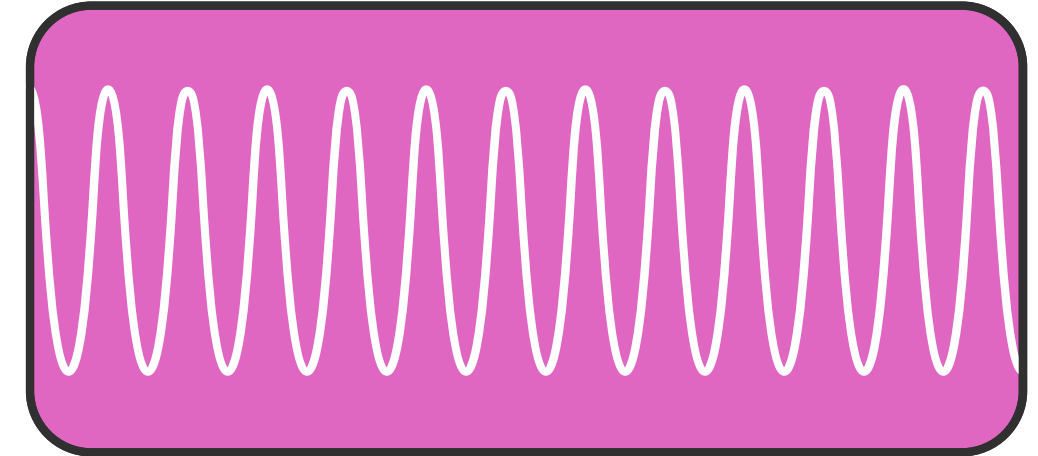


05 **ULTRAVIOLET RADIATION**

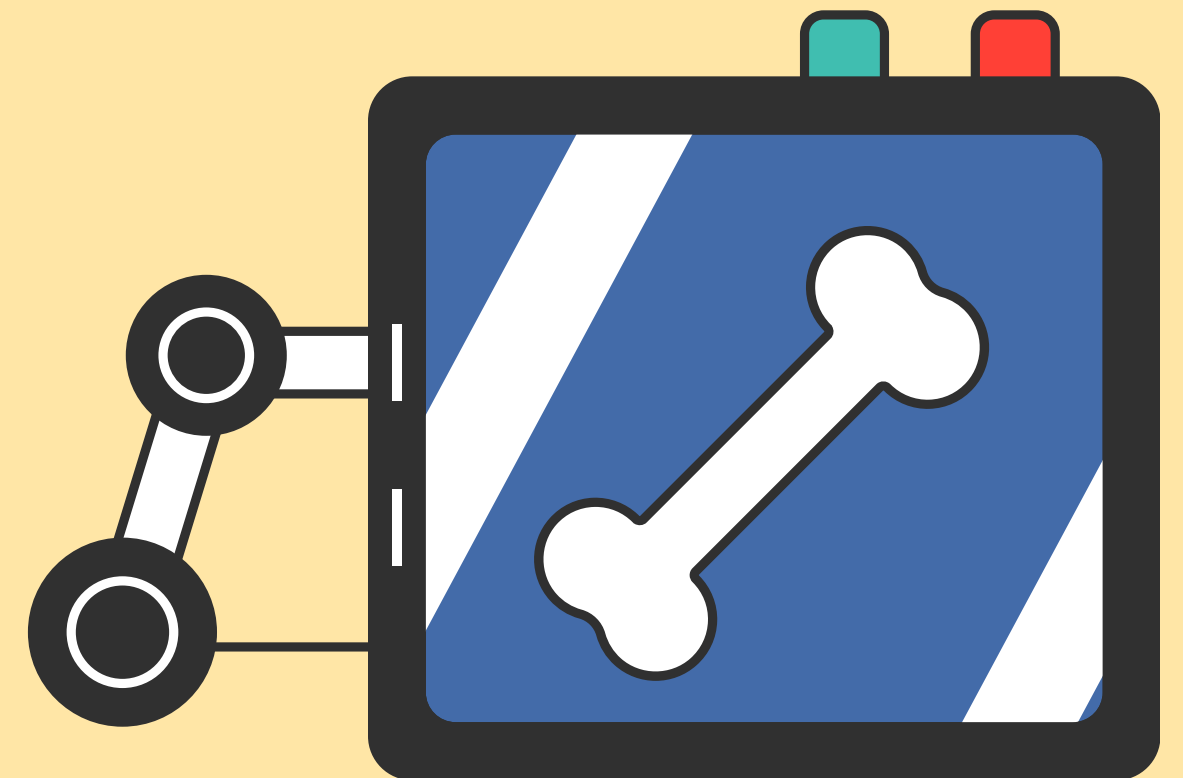


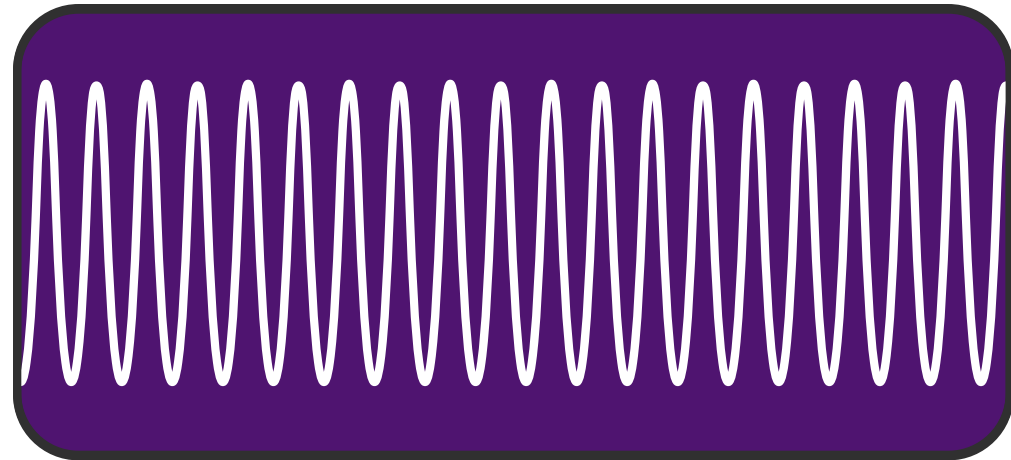
UV radiation has shorter wavelengths and higher frequencies than visible light. It comes from the sun and is responsible for causing sunburns. We can't see UV radiation, but it can be harmful to our skin and eyes.

06 X- RADIATION



X-rays have even shorter wavelengths and higher frequencies than UV radiation. They can pass through our bodies and are used in medical imaging, like X-ray machines, to see our bones and organs.

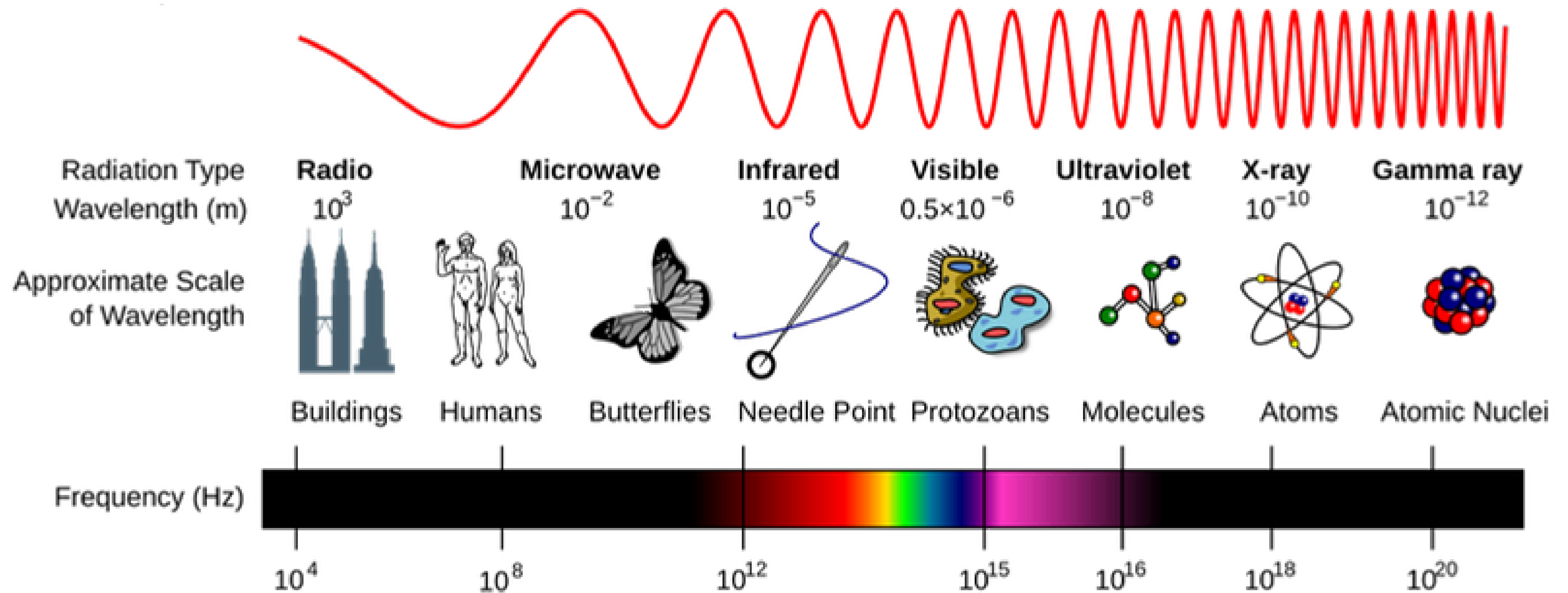




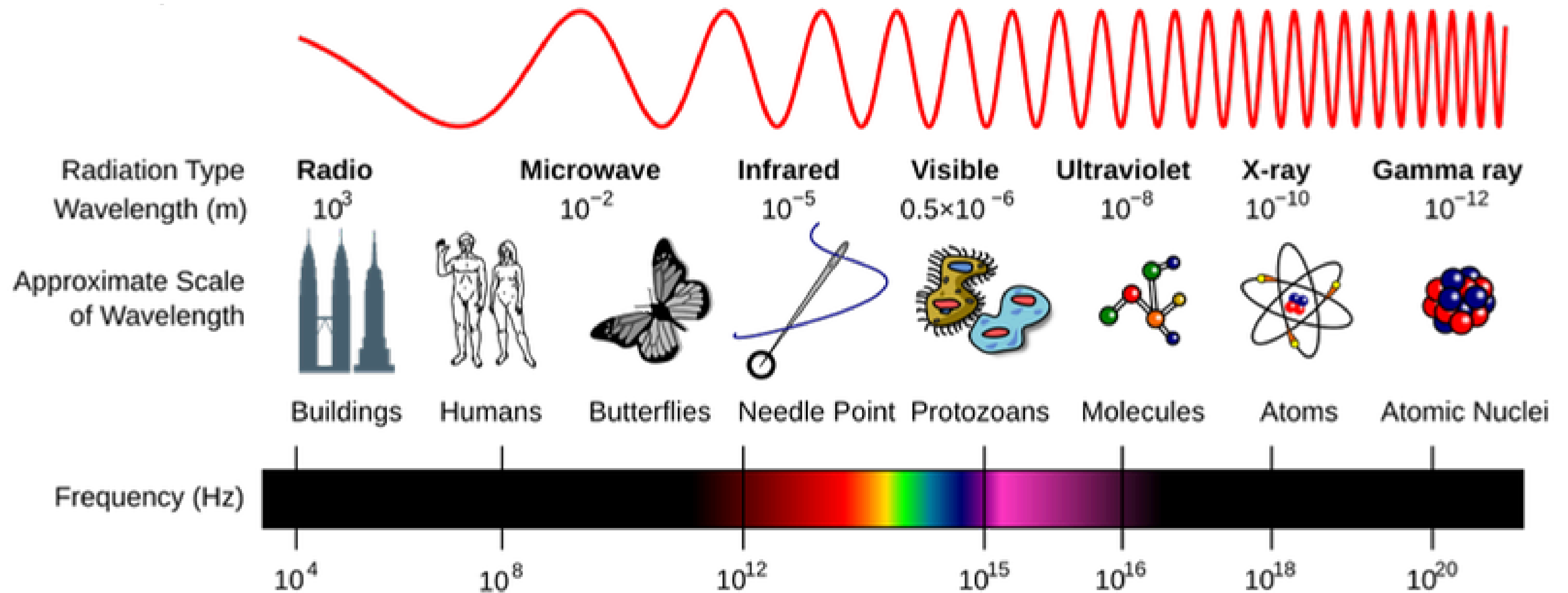
07 **GAMMA RADIATION**



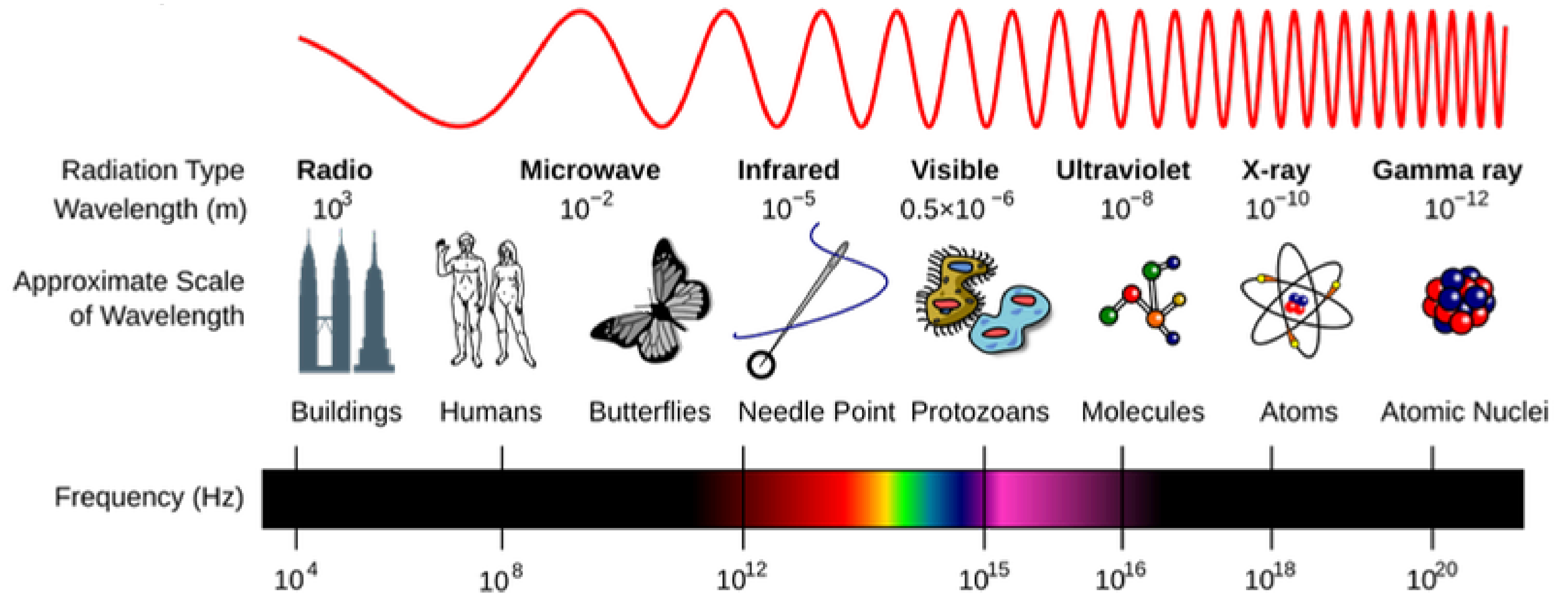
Gamma rays have the shortest wavelengths and highest frequencies of all the radiations. They are produced during nuclear reactions and can be very harmful. Scientists use gamma rays to treat cancer.



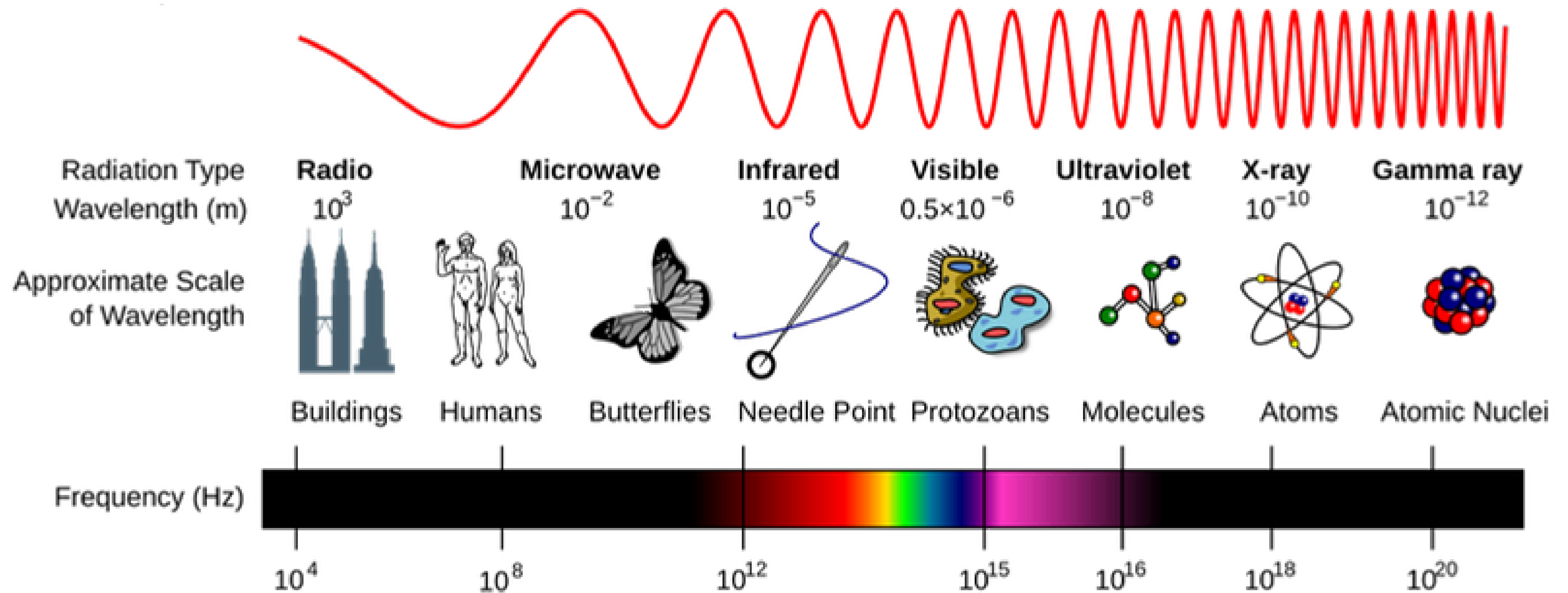
How are electromagnetic waves arranged in the electromagnetic spectrum?



What happens to the frequency as you move from radio waves to gamma rays?



What happens to the wavelength as frequency increases?



Which electromagnetic wave has the longest wavelength? Which has the highest frequency?